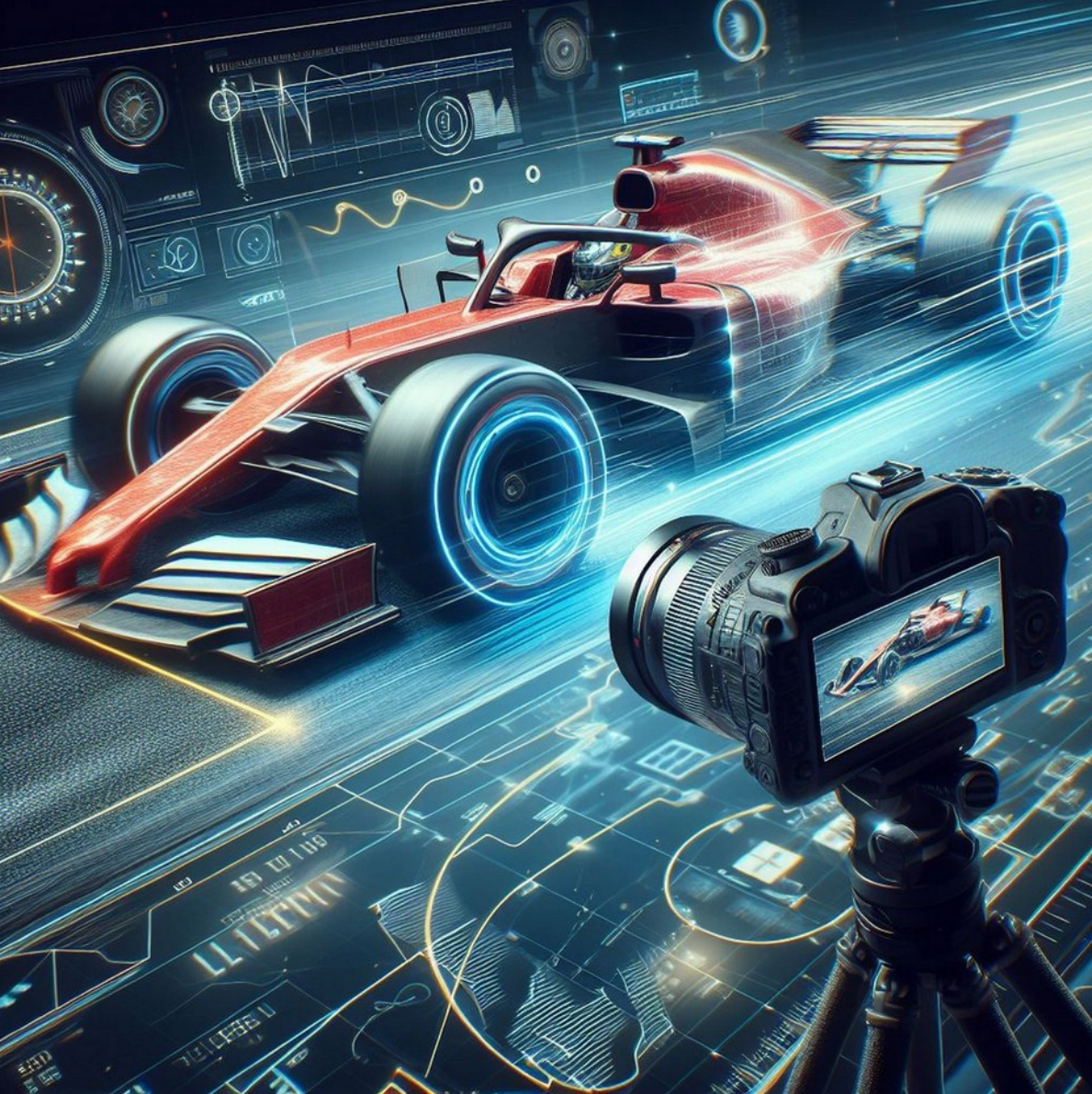


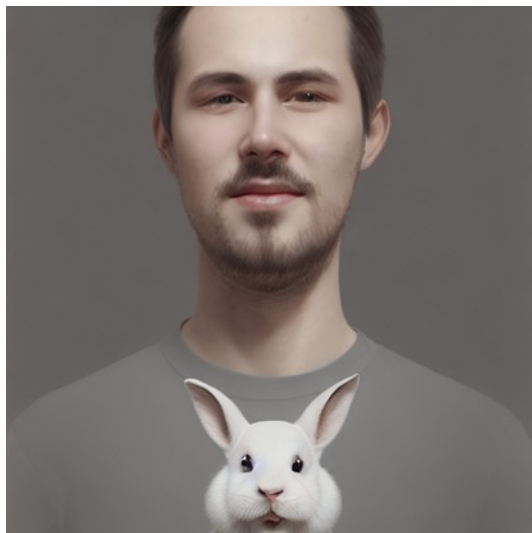


Motion Analysis

ENSIA, Algeria, May 2025

Luka Čehovin Zajc, University of Ljubljana

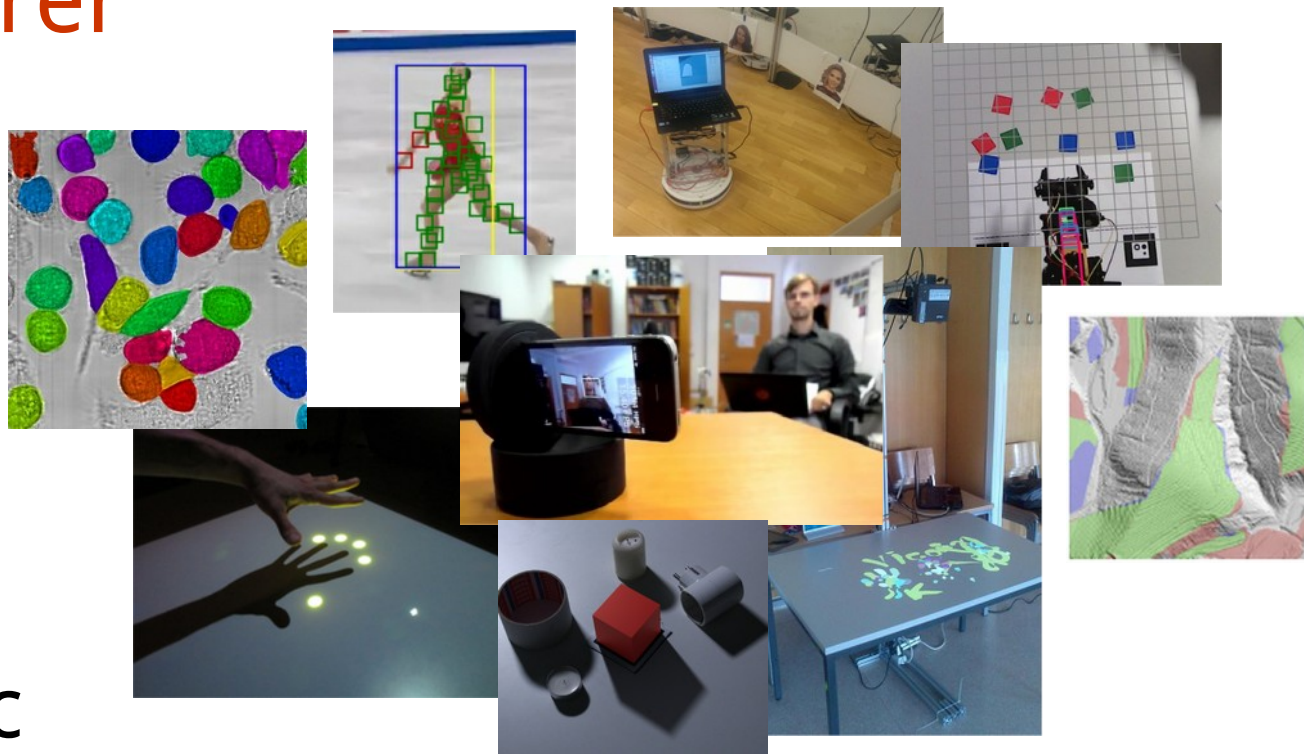
About the lecturer



Luka Čehovin Zajc

Assistant professor

University of Ljubljana, Slovenia



<https://www.vicos.si/lukacu>
<https://github.com/lukacu>



1919
established

37.500
students

6.900
employees

University of Ljubljana

23
faculties

3
art
academies



250
employees



20
research
laboratories

1996
year of
establishment

Faculty of Computer and Information Science

1973
CS becomes a
study programme
at the UL

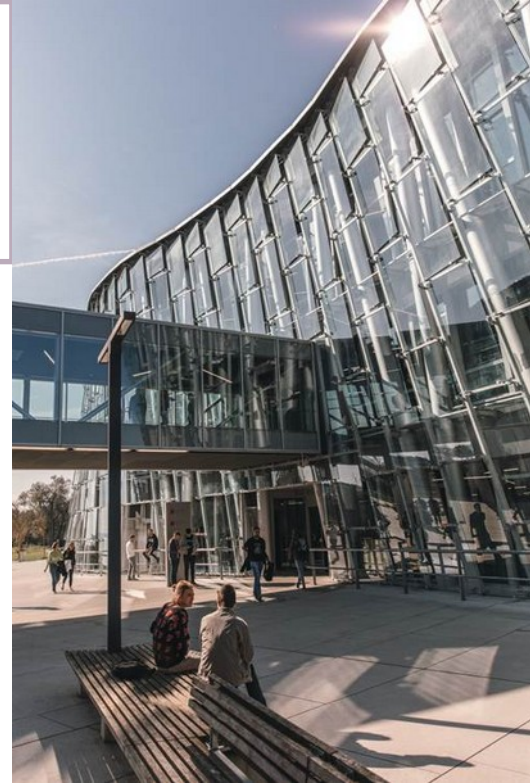
1.650
students

12
computer
classrooms

1961
beginnings of CS
research at the UL

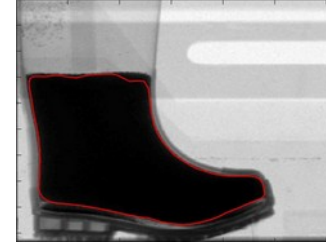
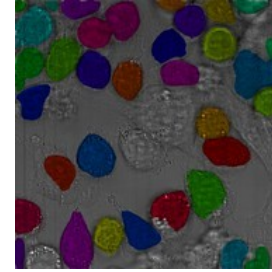
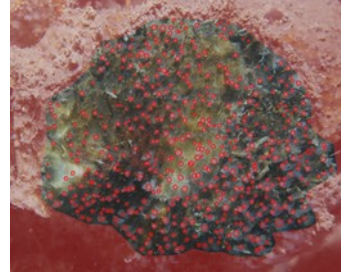


5.500
alumni



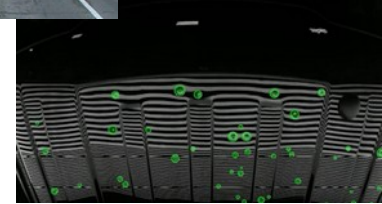
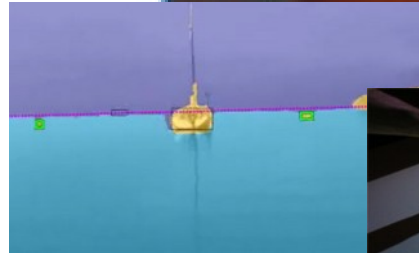
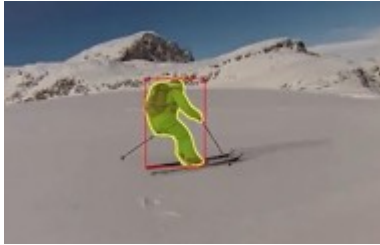
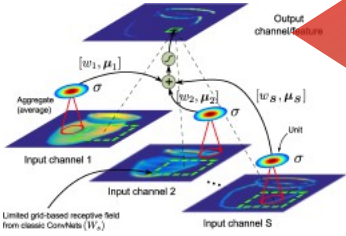
ViCoS Laboratory





Basic research

Applicative research



VOT Challenge

- International initiative
- Since 2013
- Yearly activity
 - Challenges
 - Workshops



Motion analysis in video



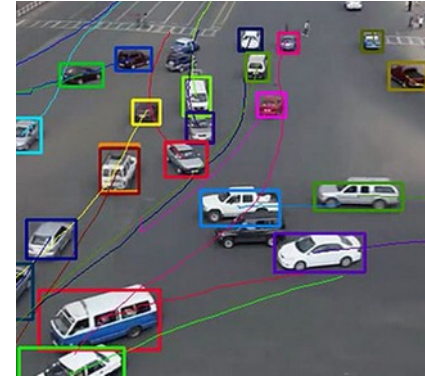
Optical-flow

Image pairs, all pixels



Point tracking

Many frames, many points



Object tracking

Entire video, Entire objects

Optical Flow

- Perceived motion in pixel (x,y) at time t
- Basic building block
 - Video compression
 - Object tracking
 - Scene understanding
 - Robotics



Problem Definition

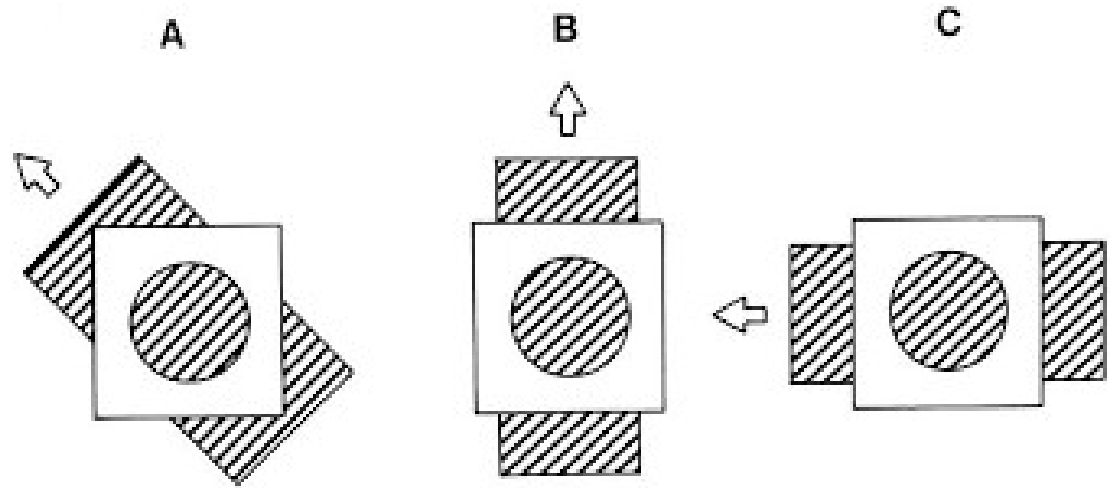
- Assumptions
 - Constant brightness
 - Small displacements
- Calculate displacements over pair of frames

$$I(x, y, t) - I(x + u, y + v, t + 1) = 0$$

$$I_x u + I_y v + I_t = 0$$

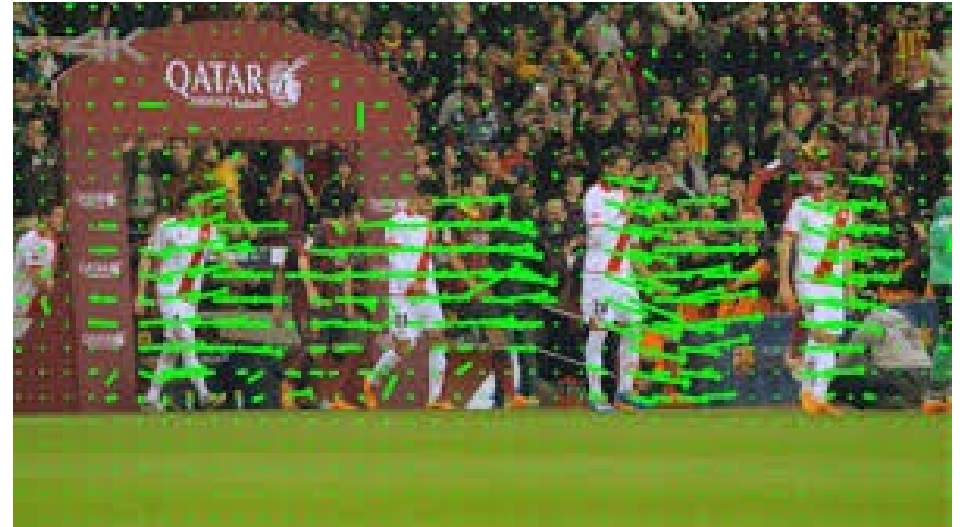
Challenges

- Projection of 3D motion to image
- Aperture problem



Lucas-Kanade Method

- Assumes constant velocity in a local patch
- Considers nearby pixels (least squares)



Lucas-Kanade Algorithm

- Numerically compute derivatives
- Solve system of equations

$$I_x(q_1)V_x + I_y(q_1)V_y = -I_t(q_1)$$

$$I_x(q_2)V_x + I_y(q_2)V_y = -I_t(q_2)$$

$$\vdots$$

$$I_x(q_n)V_x + I_y(q_n)V_y = -I_t(q_n)$$

$$\begin{bmatrix} V_x \\ V_y \end{bmatrix} = \begin{bmatrix} \sum_i I_x(q_i)^2 & \sum_i I_x(q_i)I_y(q_i) \\ \sum_i I_y(q_i)I_x(q_i) & \sum_i I_y(q_i)^2 \end{bmatrix}^{-1} \begin{bmatrix} -\sum_i I_x(q_i)I_t(q_i) \\ -\sum_i I_y(q_i)I_t(q_i) \end{bmatrix}$$

Horn-Schunck Method

- Global smoothness constraint

$$E = \iint \left[\underbrace{(I_x u + I_y v + I_t)^2}_{\text{Brightness constancy}} + \alpha^2 \underbrace{(\|\nabla u\|^2 + \|\nabla v\|^2)}_{\text{Smoothness}} \right] dx dy$$

Horn-Schunck



Lucas-Kanade



Horn-Schunk Algorithm

- Iterative optimization

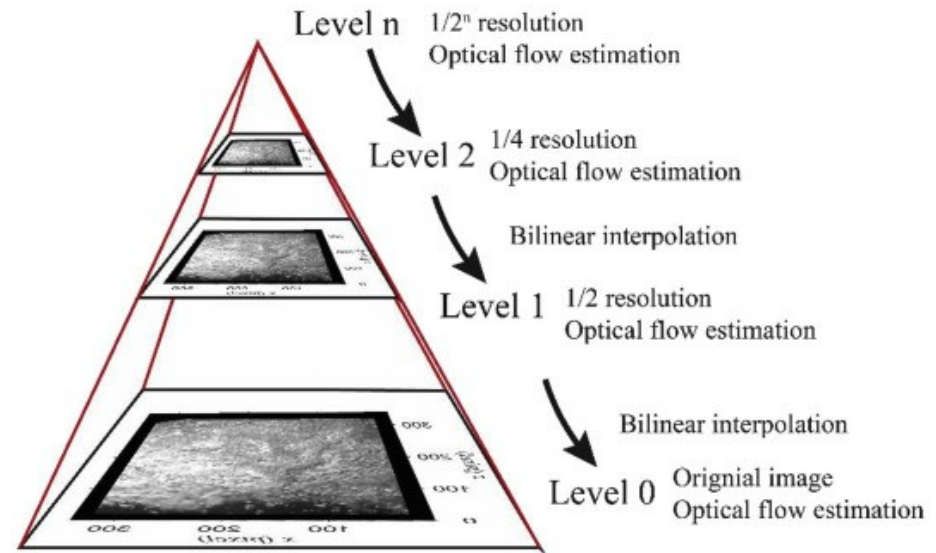
compute u'_i and v'_i as local averages of u_i, v_i

$$update = \frac{I_x u'_i + I_y v'_i + I_t}{\alpha^2 + I_x^2 + I_y^2}$$

$$u_{i+1} = u'_i - I_x update; v_{i+1} = v'_i - I_y update$$

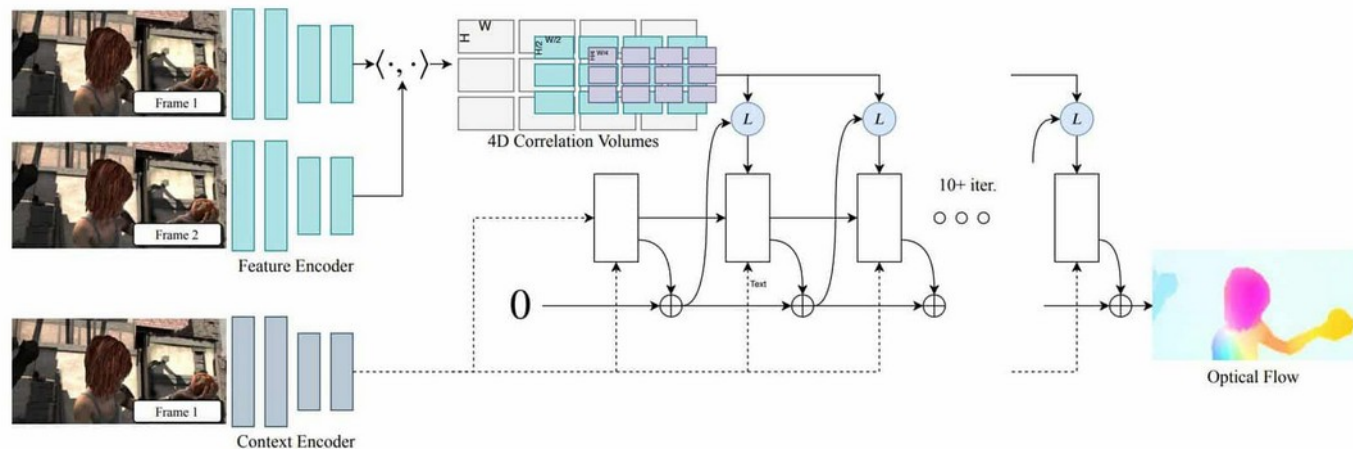
Pyramidal Optical Flow

- Handling larger displacements
- Coarse-to-fine strategy
 - Compute flow
 - Upscale and warp



Deep learning for optical flow

- Train on synthetic data
- FlowNet, RAFT

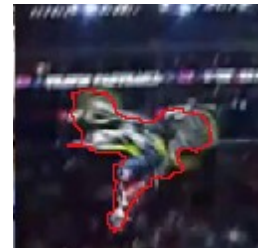
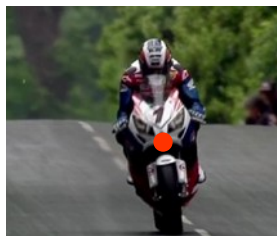


Summary

Feature	Lucas-Kanade	Horn-Schunck	Deep Learning
Motion Assumption	Constant in a small window	Smooth flow across image	Learns motion from data
Large Motion Handling	Weak (better with pyramid)	Weak (better with pyramid)	Strong with architectural design
Speed	Fast (real-time)	Moderate	Varies
Noise Robustness	Low to moderate	Moderate	High
Occlusion Handling	Poor	Poor	Data-dependent
Generalization	Low (tuned for conditions)	Assumes smoothness	Data-dependent

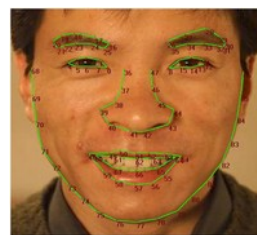
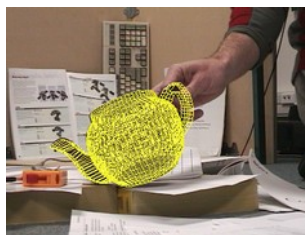
Visual Object Tracking

Predicting **state** of an object in a stream of images



Model-free tracking

center, bounding-box, segmentation

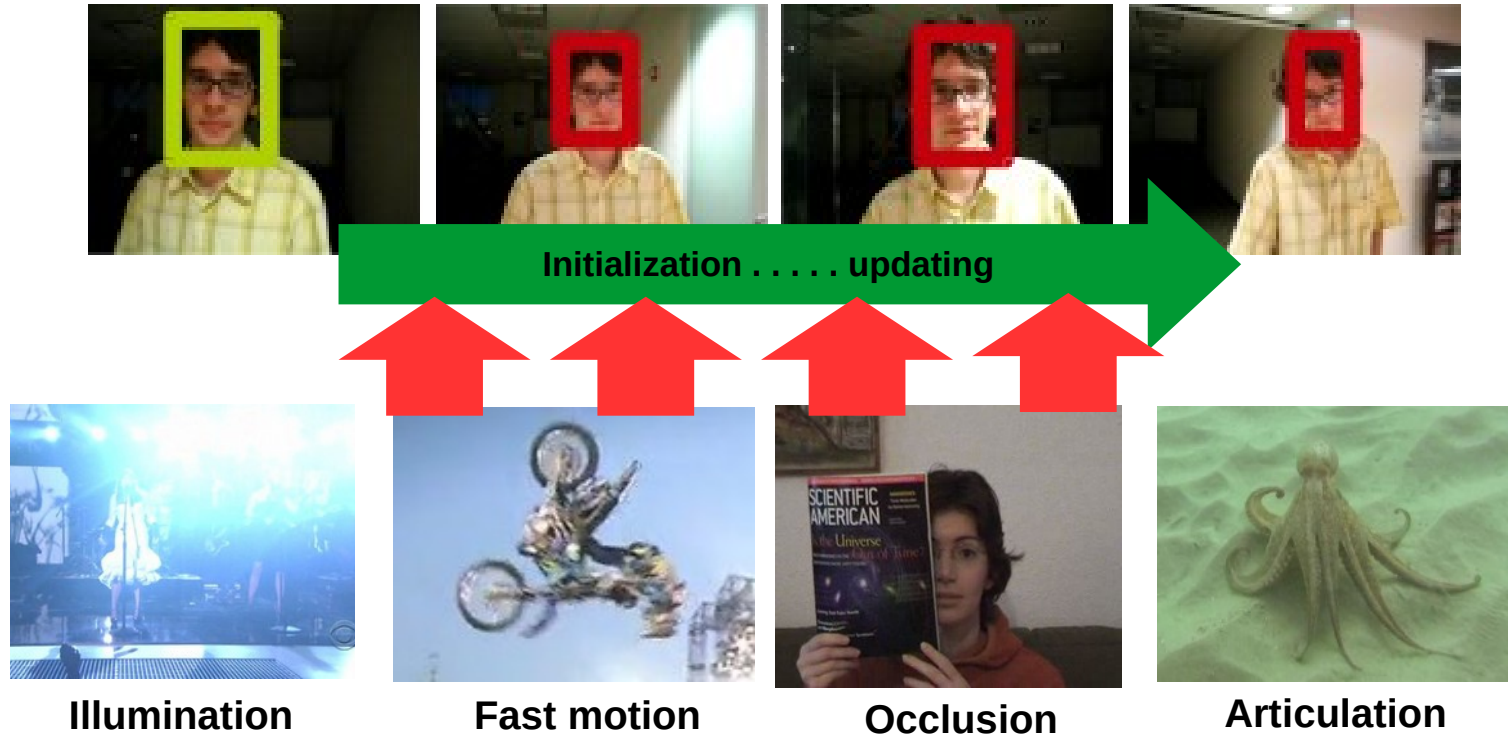


Model-based tracking

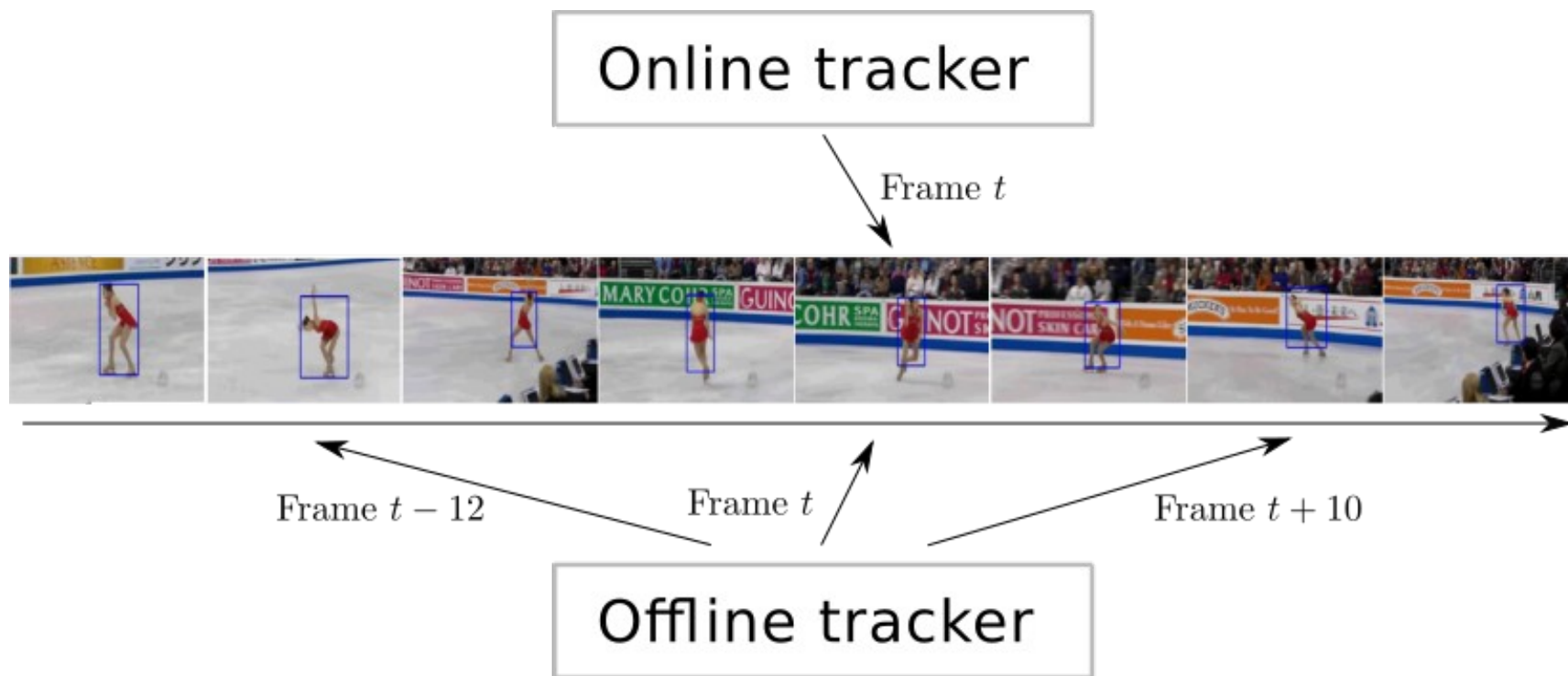
6 DoF, N DoF

Visual object tracking

Predicting state of an object in a **stream of images**

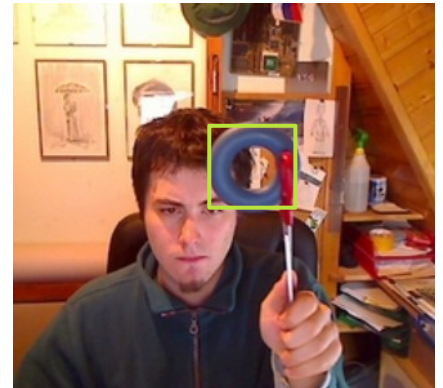
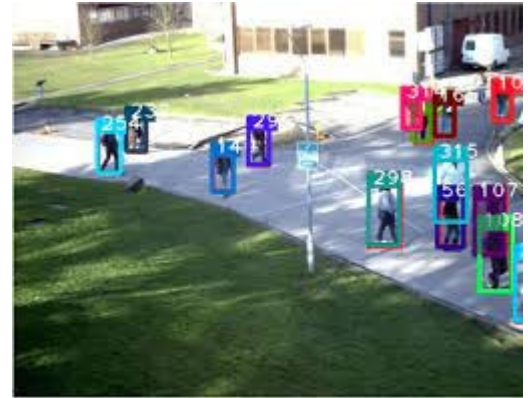


Online vs. offline



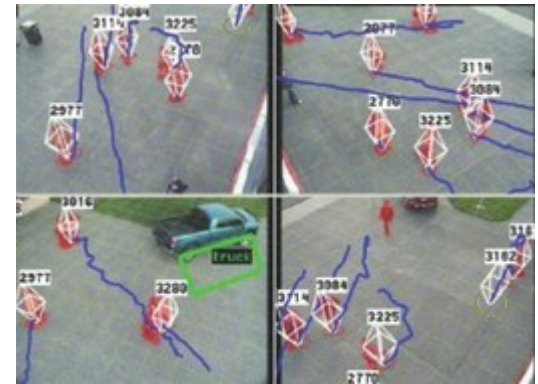
One object / many objects?

- Multi-object tracking (MOT)
 - Maintain object identity
 - Known object class?
 - Surveillance / Autonomous driving
- Single-object tracking
 - No appearance prior
 - Generality
 - Difficult conditions



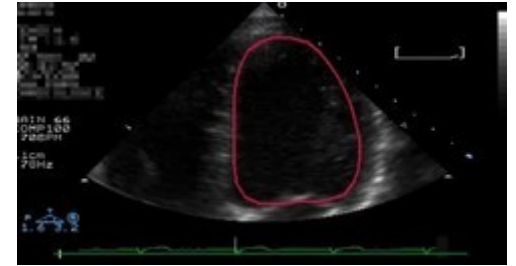
Single / multi object tracking

- Single-object tracking
 - short-term tracking
 - long-term tracking
- Multi-object tracking
 - typically long-term
 - objects often very similar
 - multi-camera



Applications

- monitoring, assistance, surveillance, control, defense
- robotics, autonomous car driving, rescue
- measurements: medicine, sport, biology, meteorology
- human computer interaction
- augmented reality
- film production and post-production: motion capture, editing
- management of video content: indexing, search
- action and activity recognition
- image stabilization
- emotion analysis
- camera "tracking"

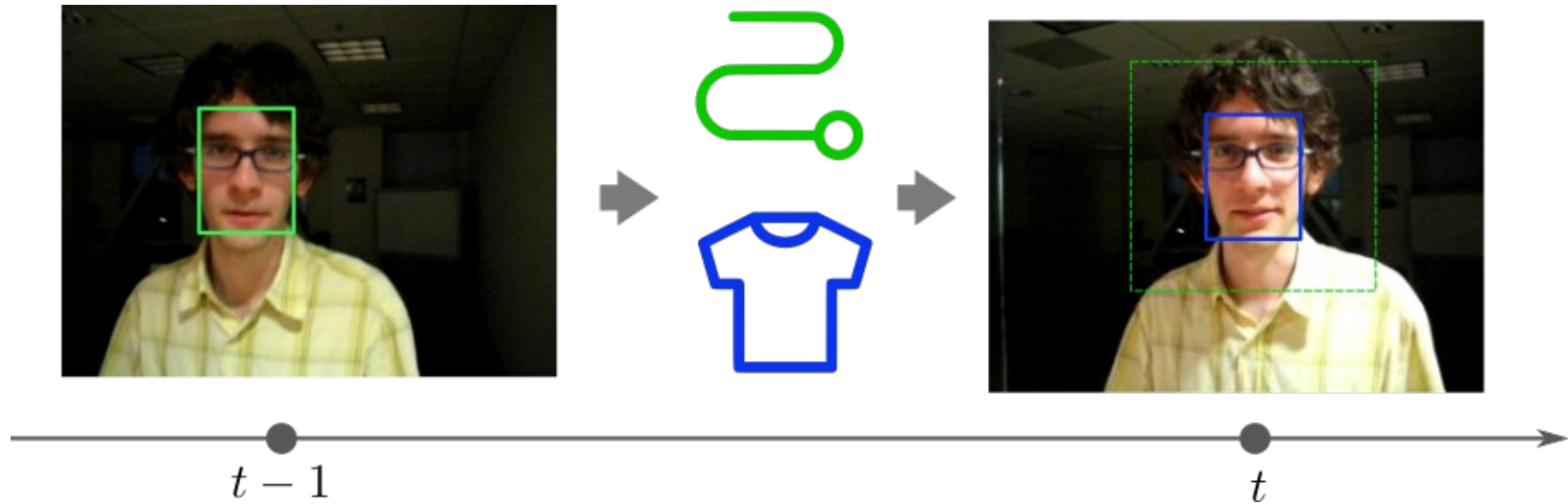


Is tracking hard?



Tracking by finding best match

- Where to look
- What to compare



The models in object tracking



Appearance

How does the object look like

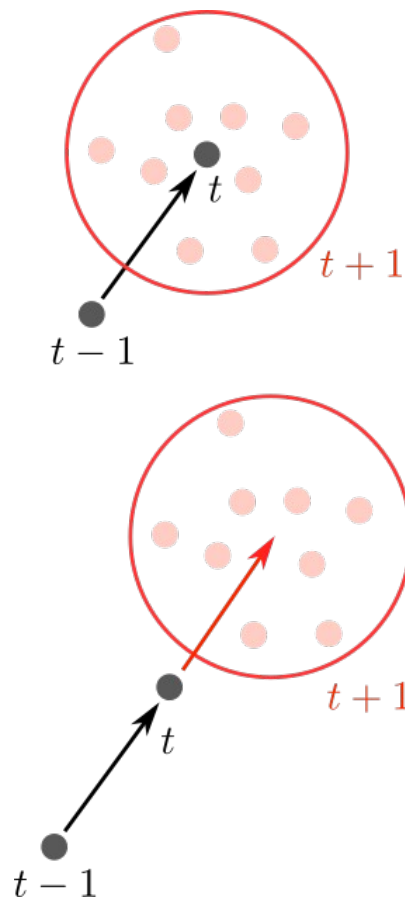


Motion

How is the object moving

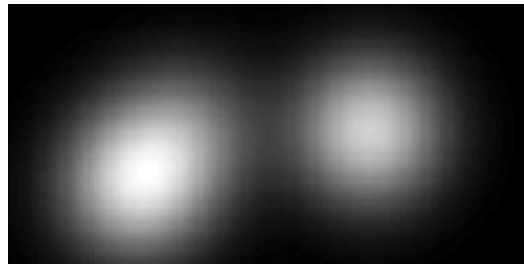
Motion models

- Predict where the object will be
- Based on previous locations
 - Random walk (Brownian)
 - Nearly constant velocity
- Integration with appearance
- Uncertainty



Recursive Bayes filters

- Combine prior knowledge, dynamic model and observations
- Uncertainty / noise modeling
- Kalman filter – Gaussian distribution
- Particle filter – Multi-modal distributions



Measurement



Motion



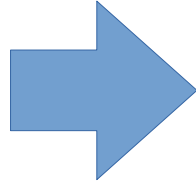
$\propto$

Posterior

Appearance model basics



Object appearance

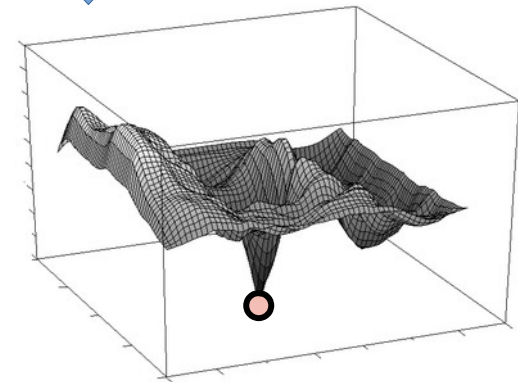
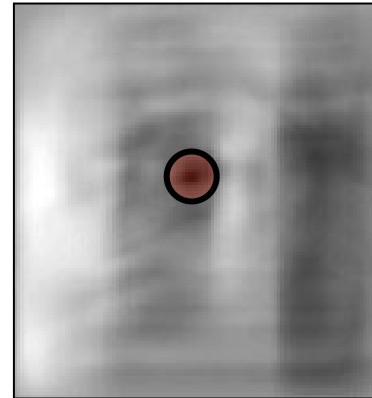
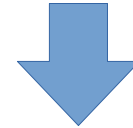


Simple matching

- Calculate the SSD for all displacements
- Select the point where similarity is maximal
- Starting position is close to the right one

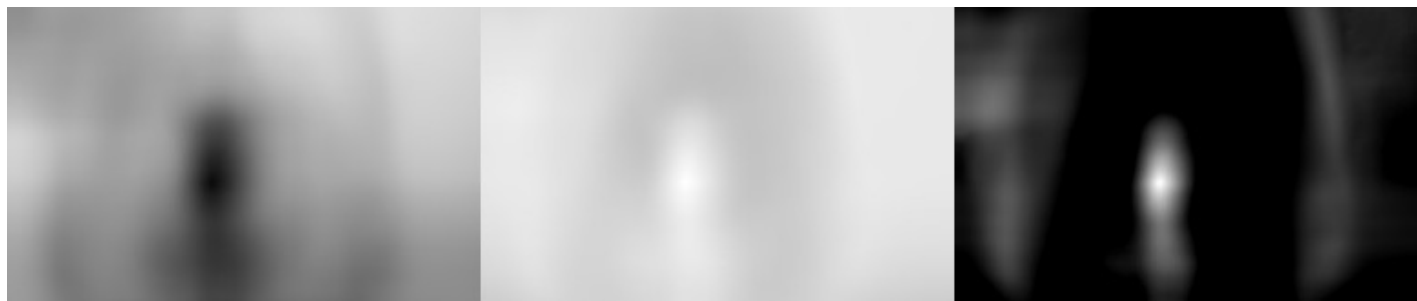
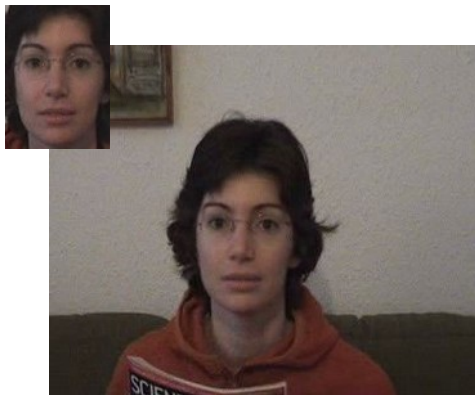


$$R(x, y) = \sum_{x', y'} (T(x', y') - I(x + x', y + y'))^2$$



Correlation matching

- Correlation $R(x, y) = \sum_{x', y'} (T(x', y') \cdot I(x + x', y + y'))$
- Normalized correlation $R(x, y) = \frac{1}{n\sigma_I\sigma_T} \sum_{x, y} (I(x, y) - \mu_I) (T(x, y) - \mu_T)$

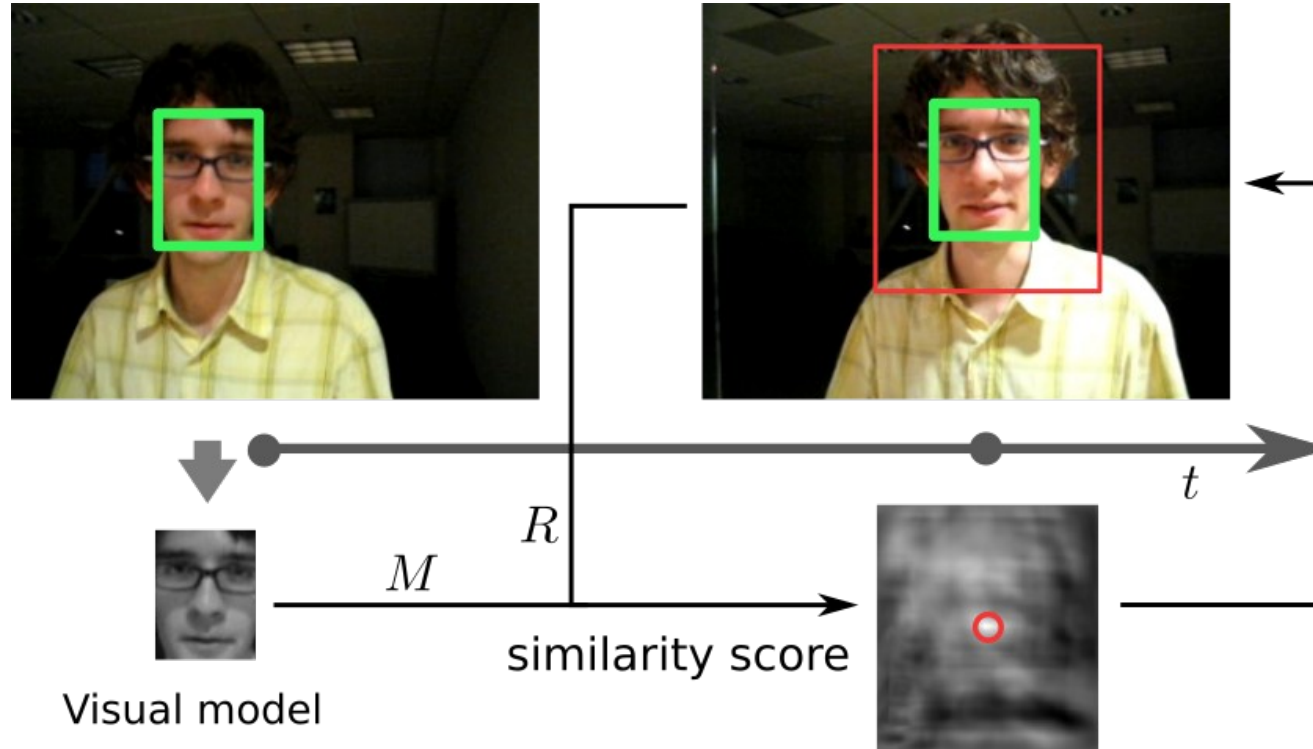


SSD

Correlation

NCC

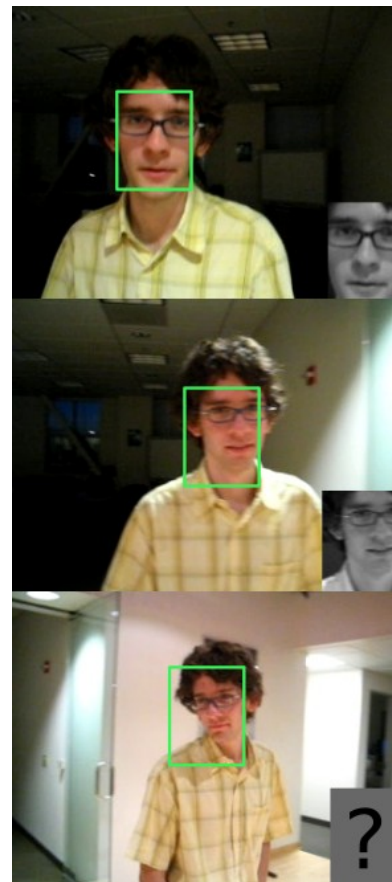
Case study: NCC





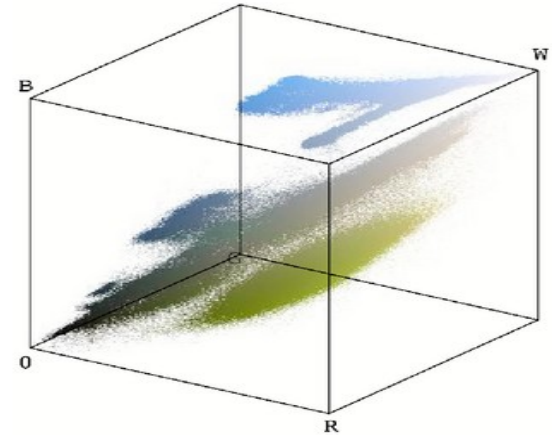
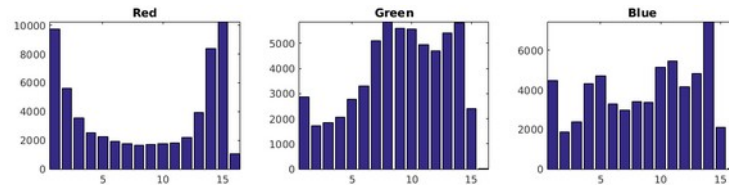
To update or not to update?

- Appearance changes
- Concept drift
- Model design goals
 - Invariance
 - Fast updates
 - Robust to drift



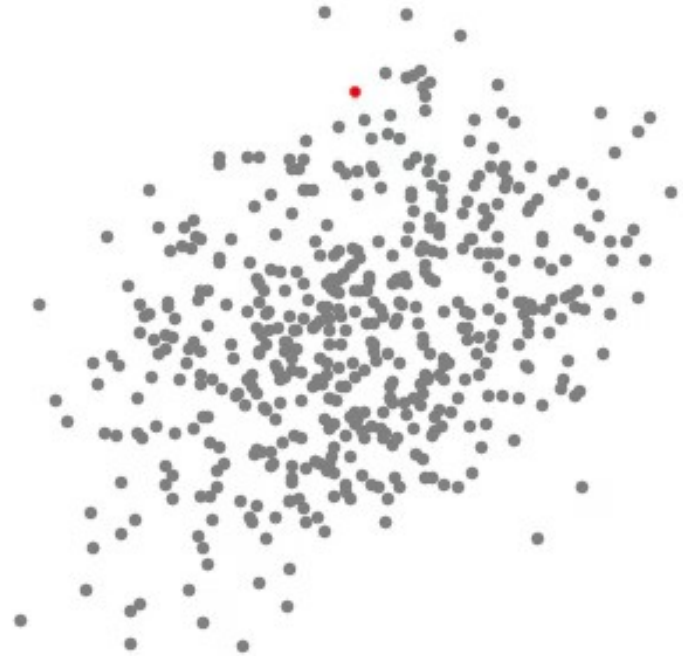
Tracking using color

- Object color is invariant
 - Geometry changes
- Color model
 - Parametric (Gaussian)
 - Non-parametric (Histogram)



Mode seeking

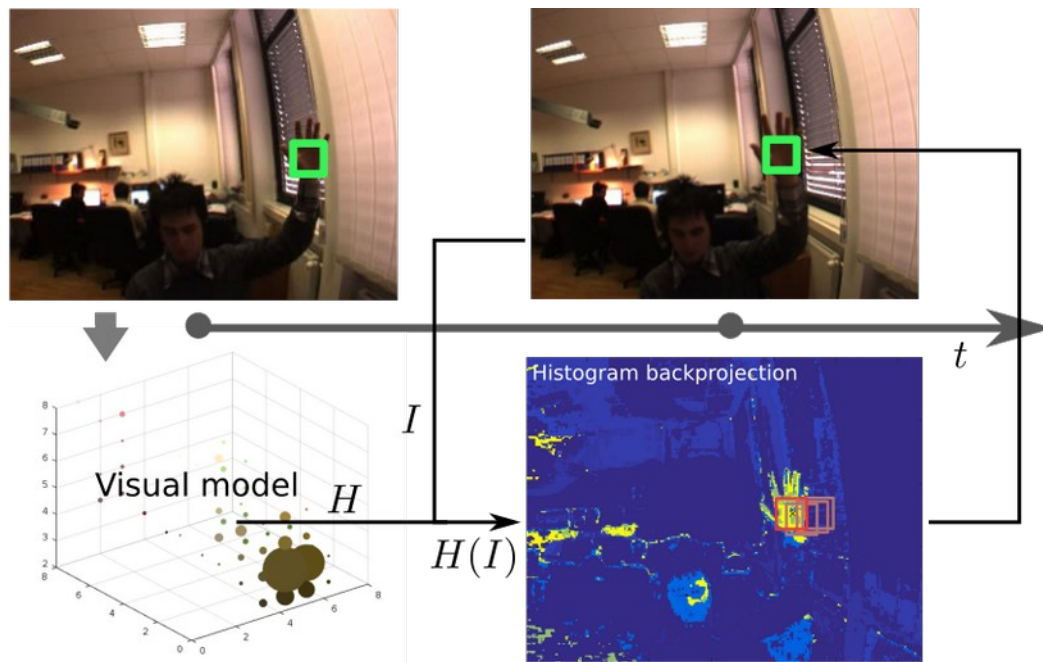
- Iterative algorithm
 - Window
 - Move to mean value
- Gradient ascent
- Mean-shift

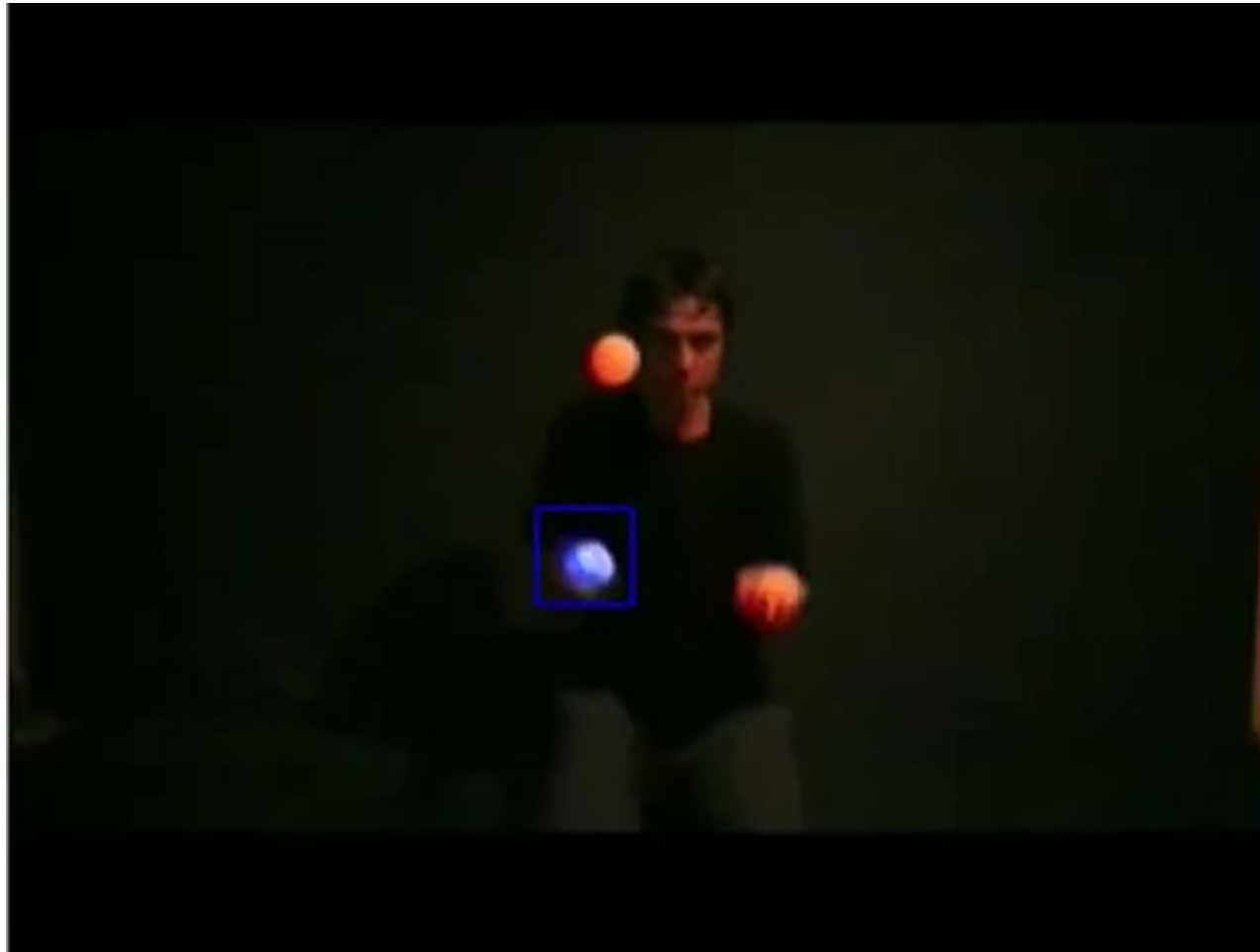


Case study: MeanShift

- Color histograms
 - Color space
 - Number of bins
 - Updating
- Bhattacharyya distance

$$\rho(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^B \sqrt{p_i \cdot q_i}$$





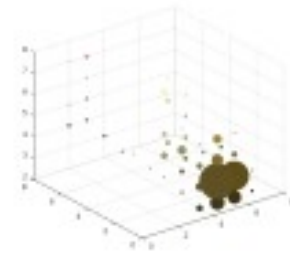
Template vs. histogram



Template

Very specific

Does not handle deformations



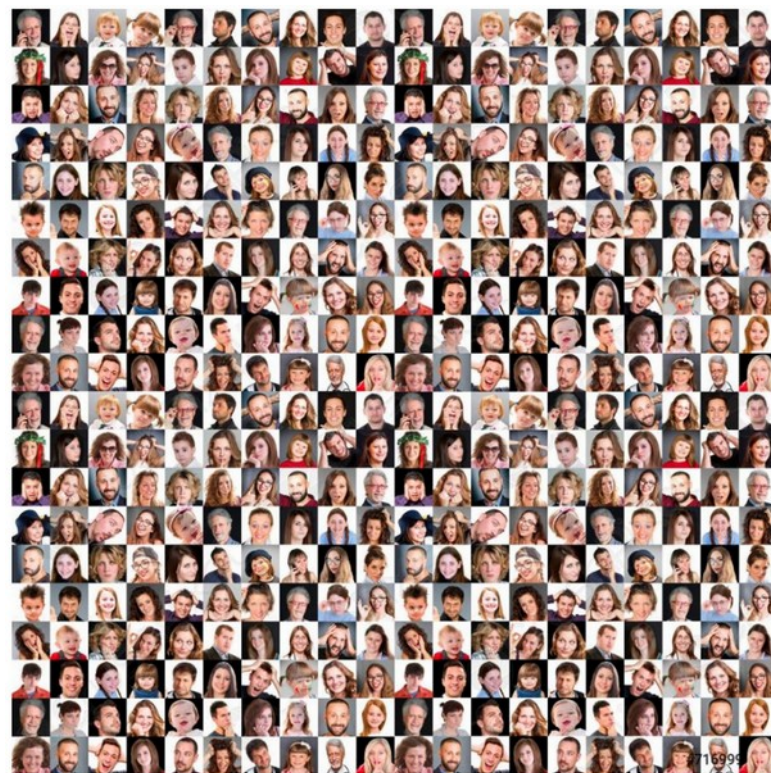
Histogram

Invariant to transformation

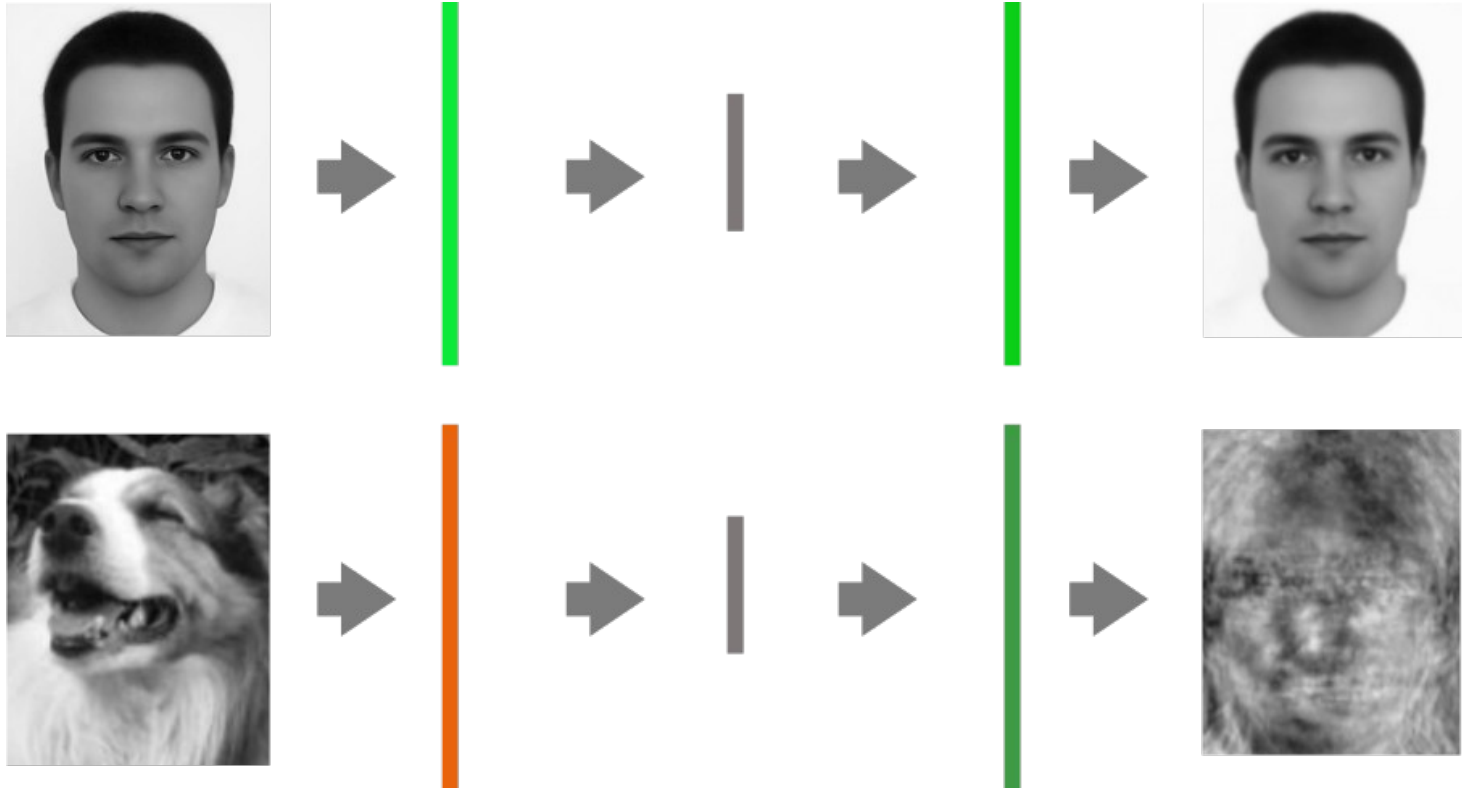
Not that specific

Template subspaces

- Image - a point in space
 - Find a linear subspace
 - Reduce information loss
- Subspaces in tracking
 - Model object appearance
 - Reconstruction as similarity

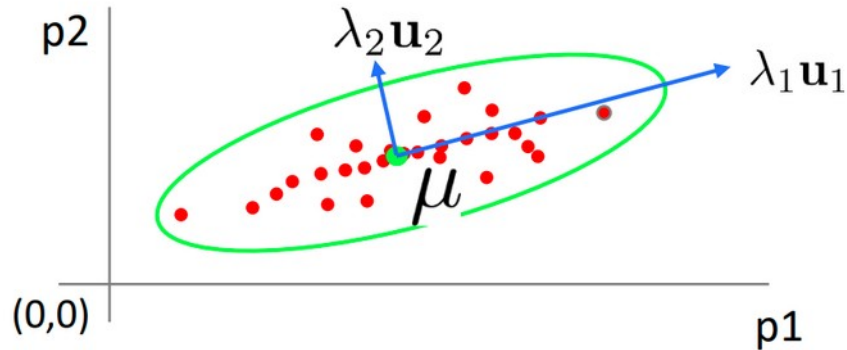


Subspace projection

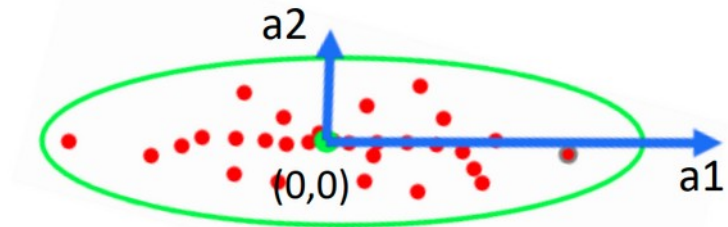


Principal Component Analysis

- Decomposition of covariance matrix
 - Eigenvectors: directions of variance
 - Eigenvalues: variance of data

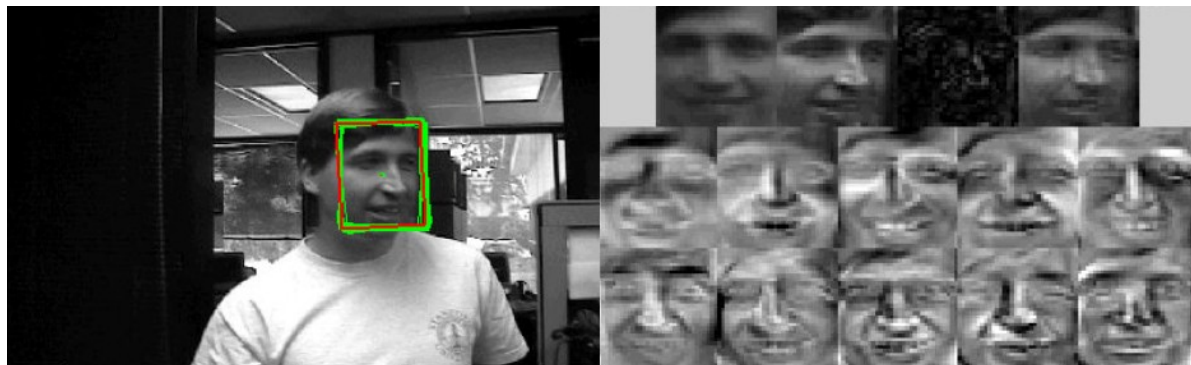


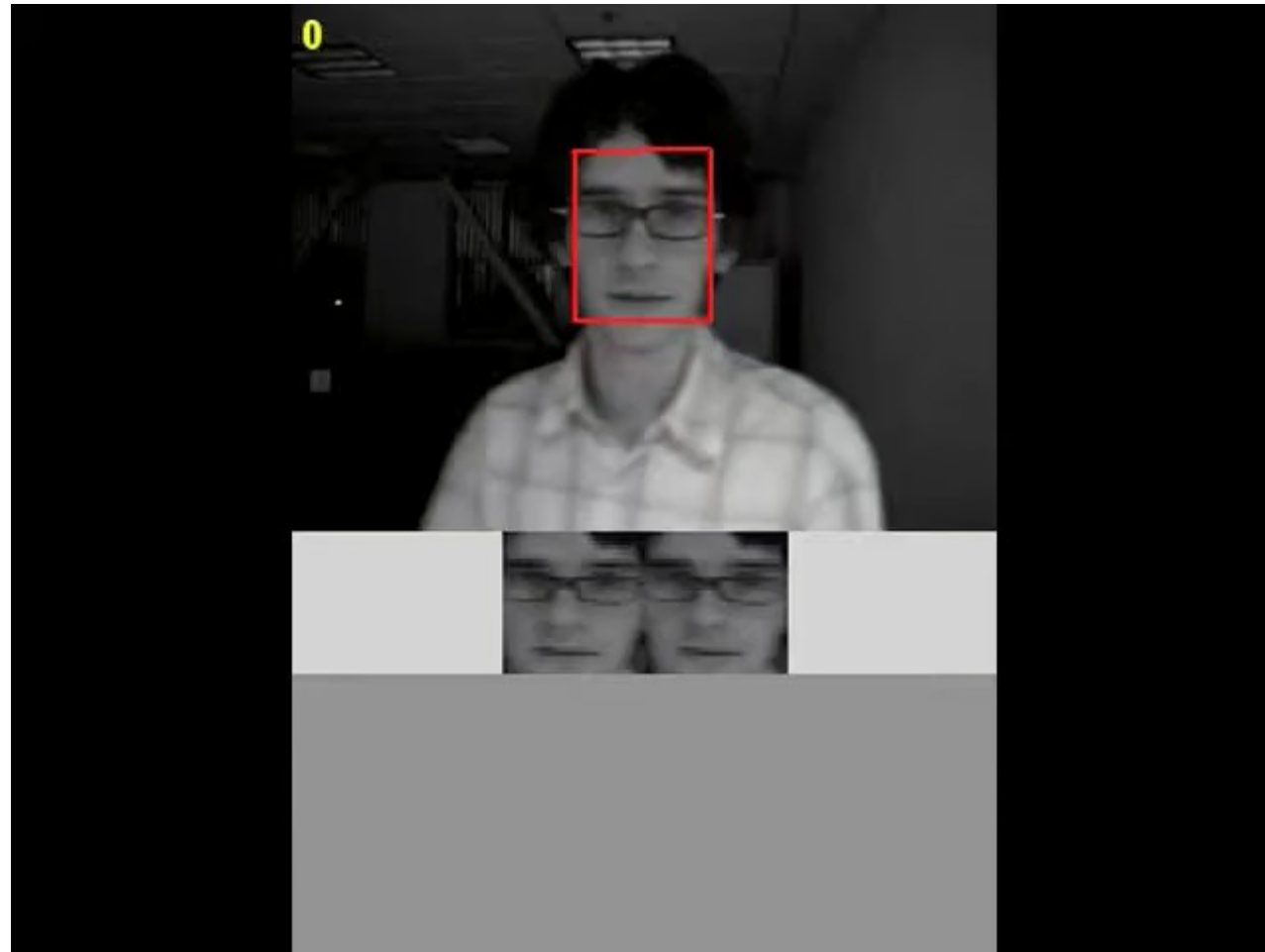
1. Translate to origin by $\mathbf{t} = \mu$
2. Rotate by $\mathbf{R} = \mathbf{U}$



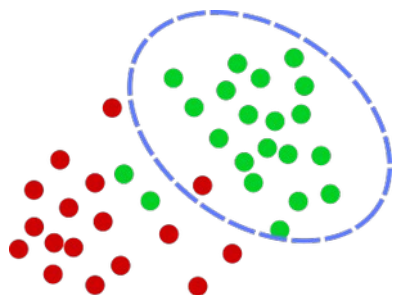
Case study: IVT

- Model possible deformations as an image subspace
 - PCA
 - Incremental updates
- Matching
 - Re-projection error
 - Sampling candidates





Discriminative tracking

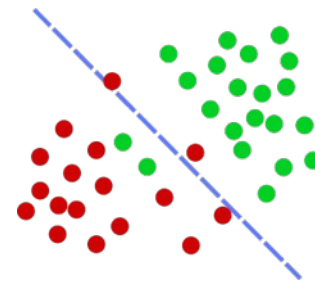


Generative model

Describing appearance

Similarity

More general



Discriminative model

Describe differences

Less data

More specific

Binary classifier as detector



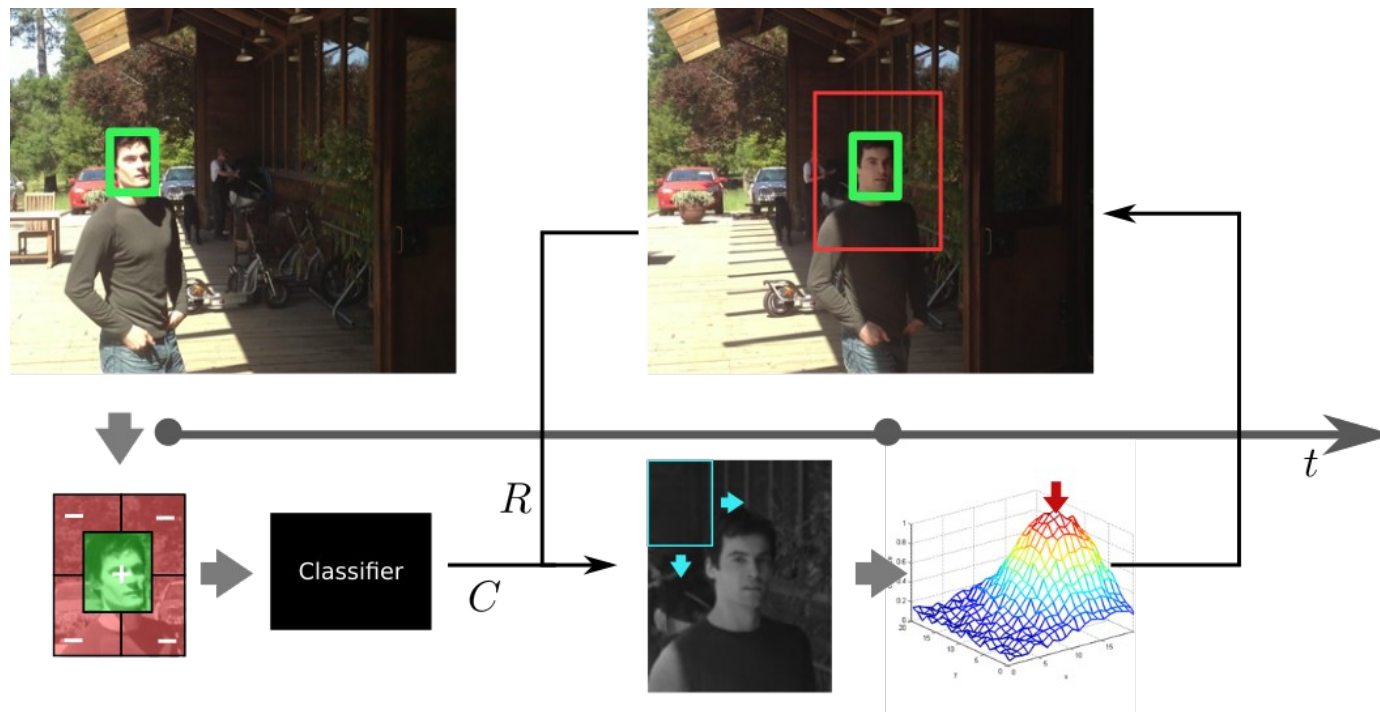
⋮



⋮

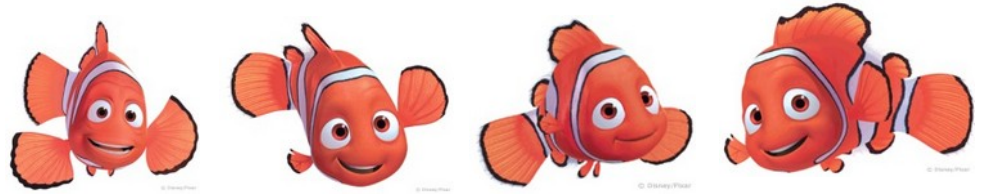


Tracking as detection



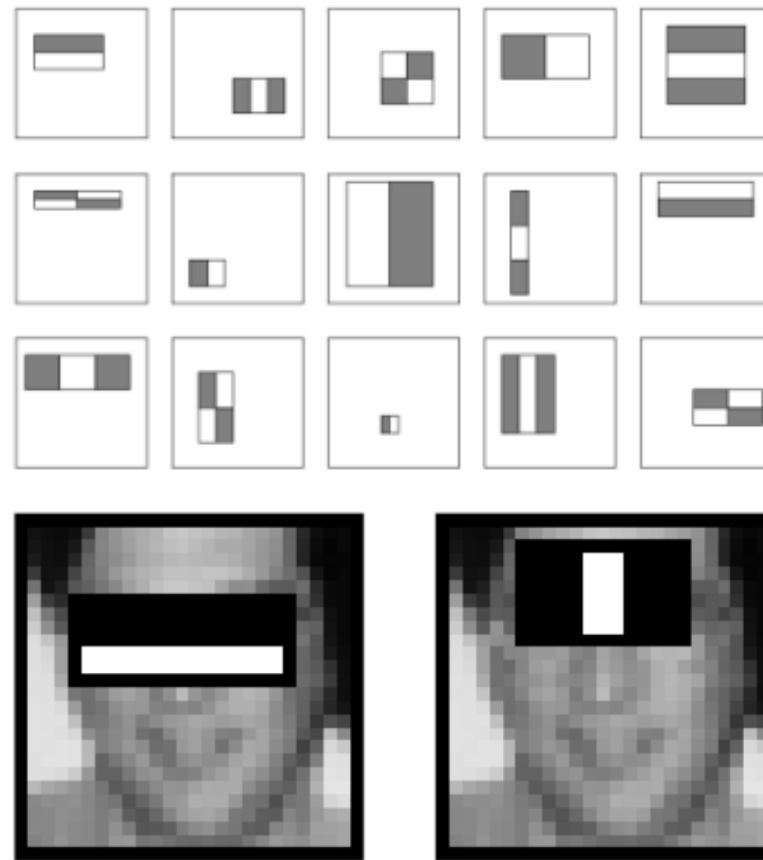
Challenges

- Adaptability
- Robustness
- Generality
- Performance
 - Training
 - Feature extraction
 - Classification



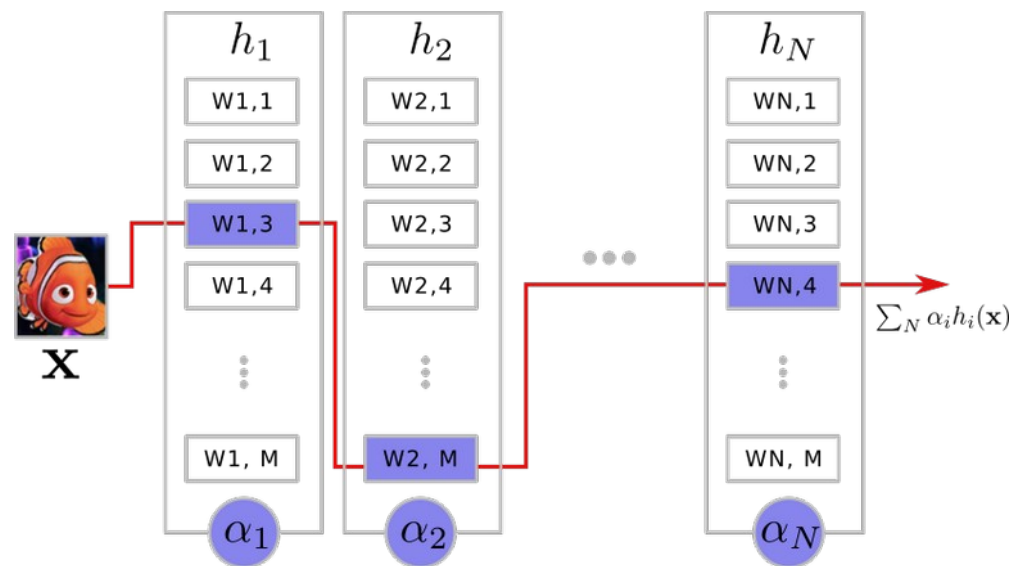
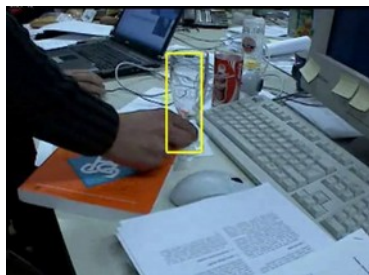
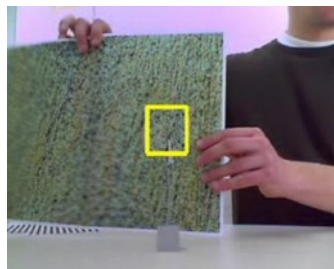
Boosting cascade

- Haar features
 - Integral image
- Cascade
 - Many negatives
 - Reject early



Boosting for tracking

- AdaBoost classifier (cascade)
- On-line updating
 - Distribution modeling
 - Classifier pools (selectors)

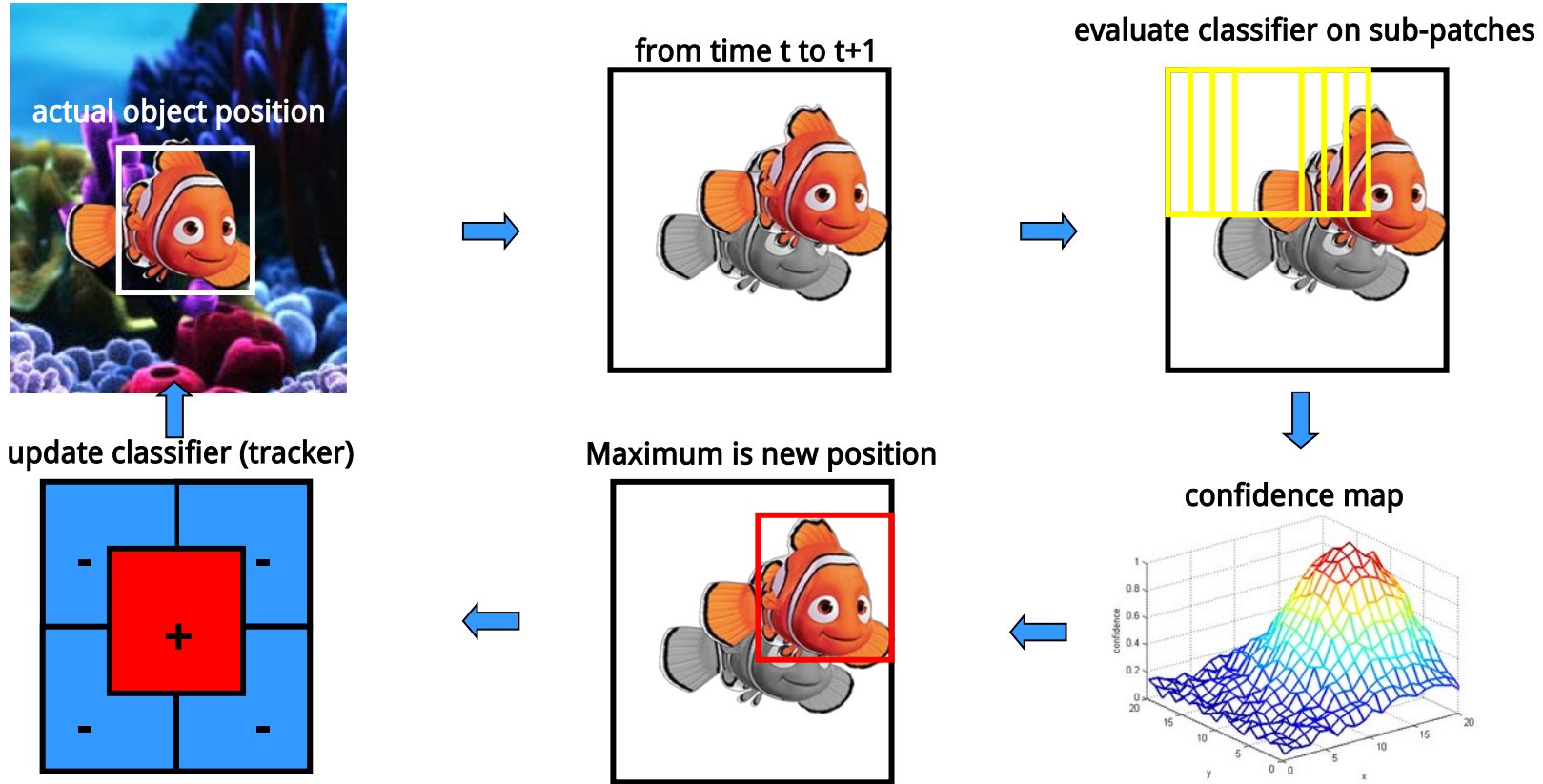


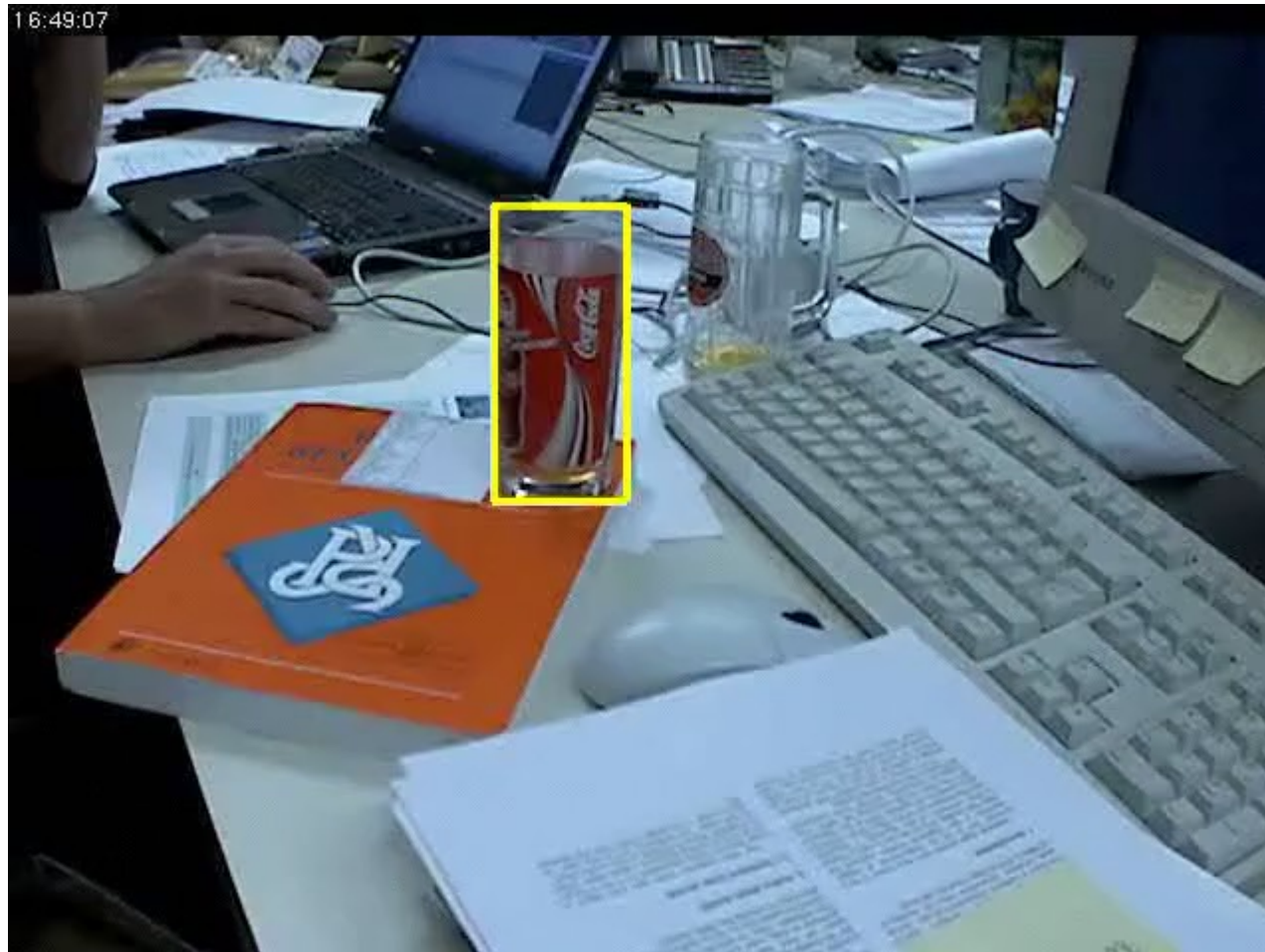
H. Grabner, H. Bischof. On-line boosting and vision. CVPR, 2006

H. Grabner, M. Grabner, H. Bischof. Real-time Tracking via Online Boosting. BMVC 2006

Babenko et al., "Robust Object Tracking with Online Multiple Instance Learning", TPAMI 2011

Case study: Online Boosting



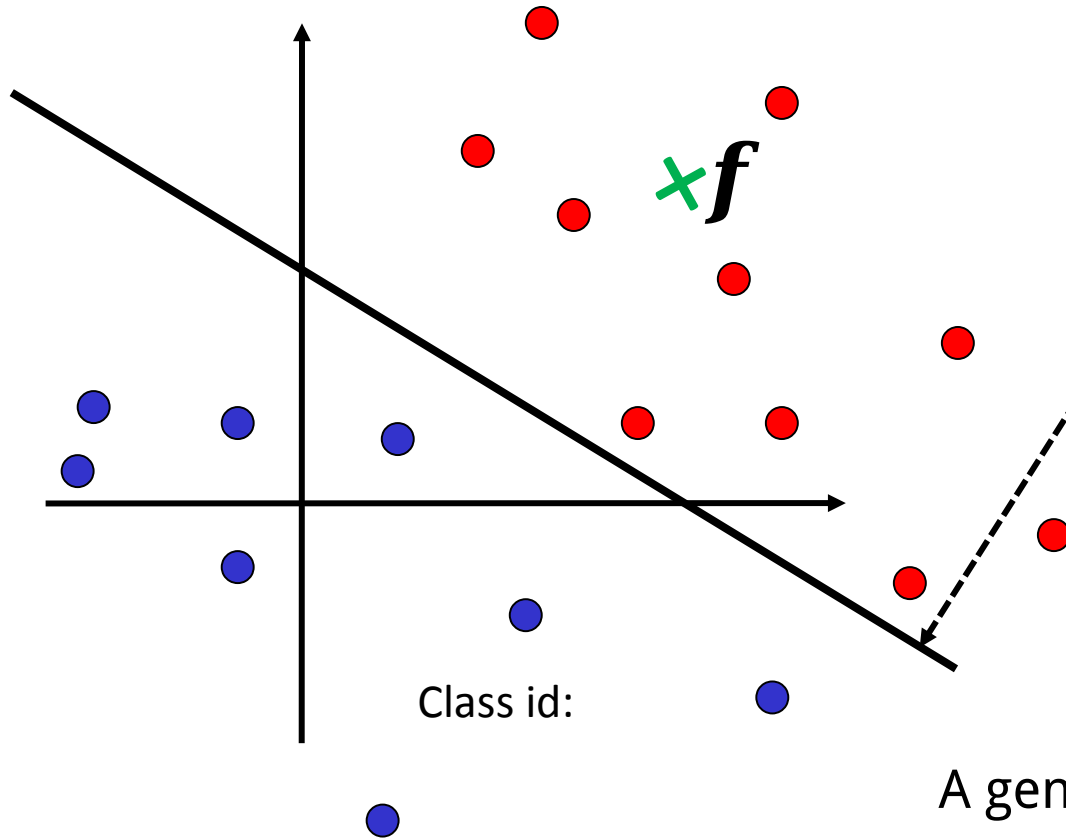


Back to the template matching

- Sliding window
- Compare with offset
- Correlation



Correlation as a linear classifier



A decision boundary, in general, a *hyper-plane*

$$af_1 + cf_2 + b = 0$$

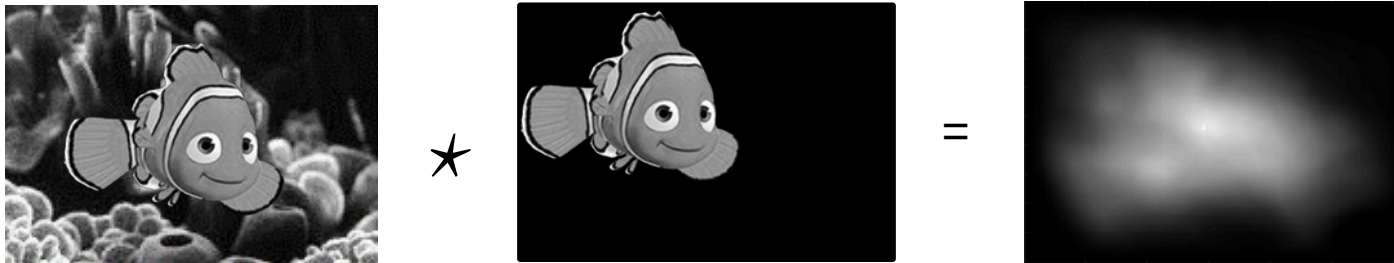
Define: $\mathbf{h} = \begin{bmatrix} a \\ c \\ b \end{bmatrix}$ $\mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ 1 \end{bmatrix}$

A general hyper-plane: $\langle \mathbf{h}, \mathbf{f} \rangle = \mathbf{h}^T \mathbf{f} = 0$

Efficient computation

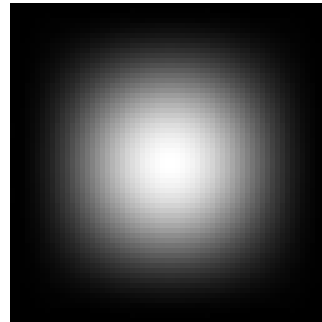
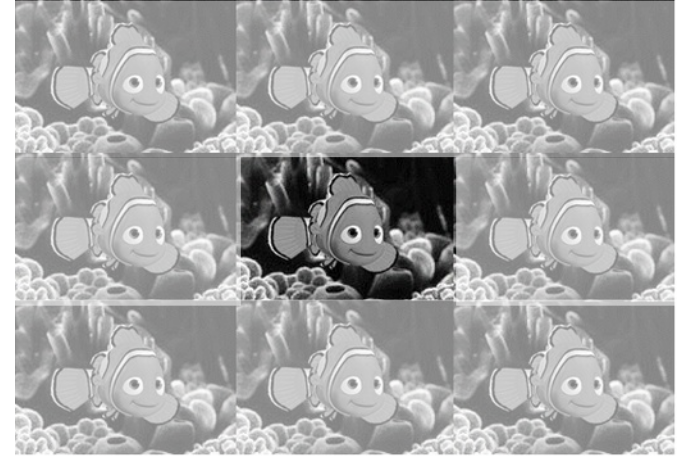
- Correlation is convolution (flipped signal)
- Point-wise product in Fourier domain
 - F and H must be of the same size

$$G = F \star H \rightarrow \hat{G} = \hat{F} \odot \overline{\hat{H}}$$

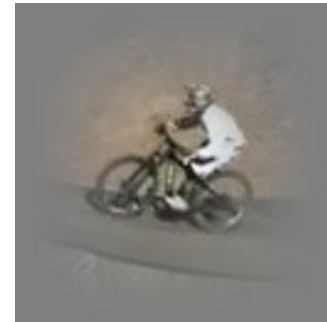


Circularity

- Correlation is circular in Fourier space
 - Boundary effect
 - Hanning window

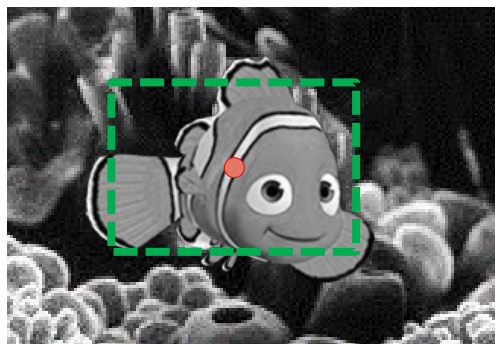


=



Tracking as correlation

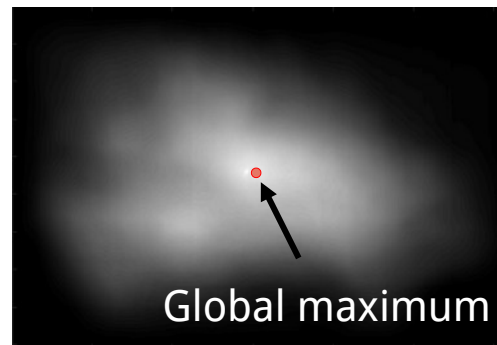
- Accelerated by FFT
 - Fast linear regressor
- Poor performance
 - Not discriminative



★



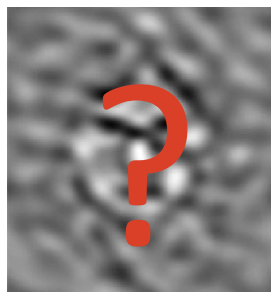
=



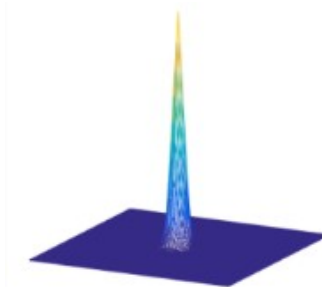
Discriminative correlation filters



Training
example



Template

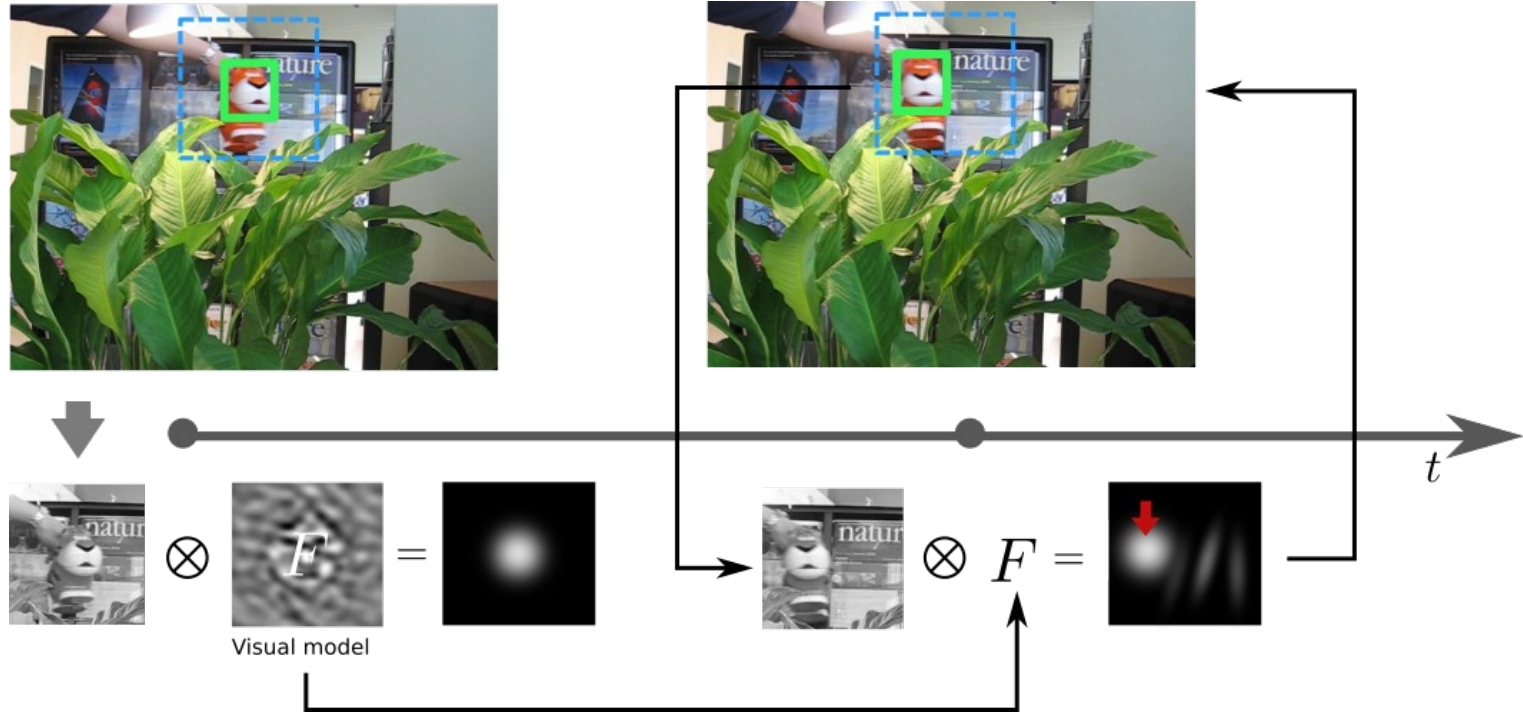


Desired
response

$$\arg \min_{\mathbf{F}} \|\mathbf{T} \star \mathbf{F} - \mathbf{G}\|^2$$

Closed-form solution
Fast computation via FFT

Case study: MOSSE





Histogram of oriented gradients

- Similar to SIFT
- Sobel filters
- Dense grid
- Block histograms
 - 16x16 pixel
 - Weighting

Input image

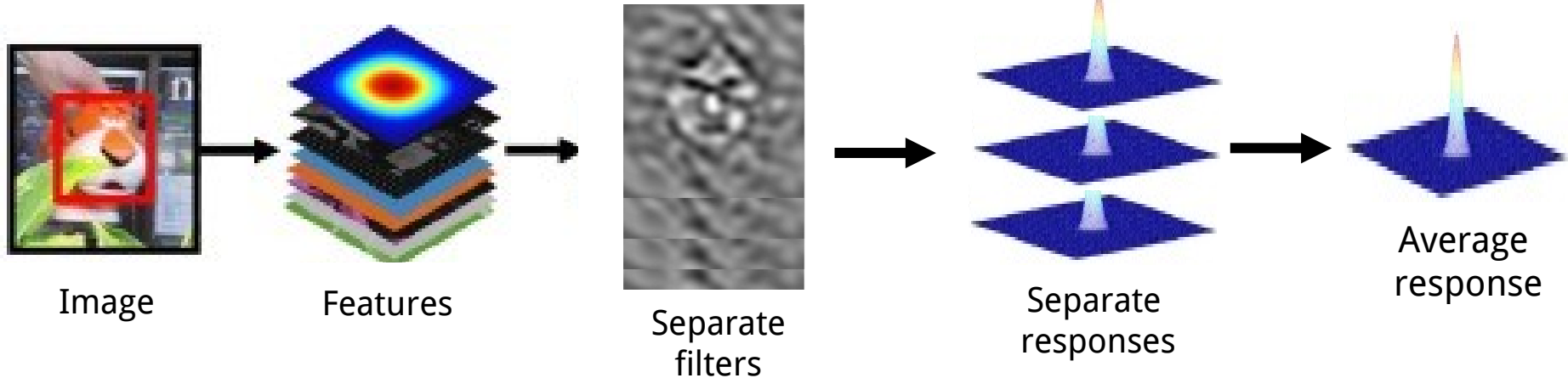


Histogram of Oriented Gradients



Case study: KCF

- Multichannel formulation (HOG)
- Scale estimation



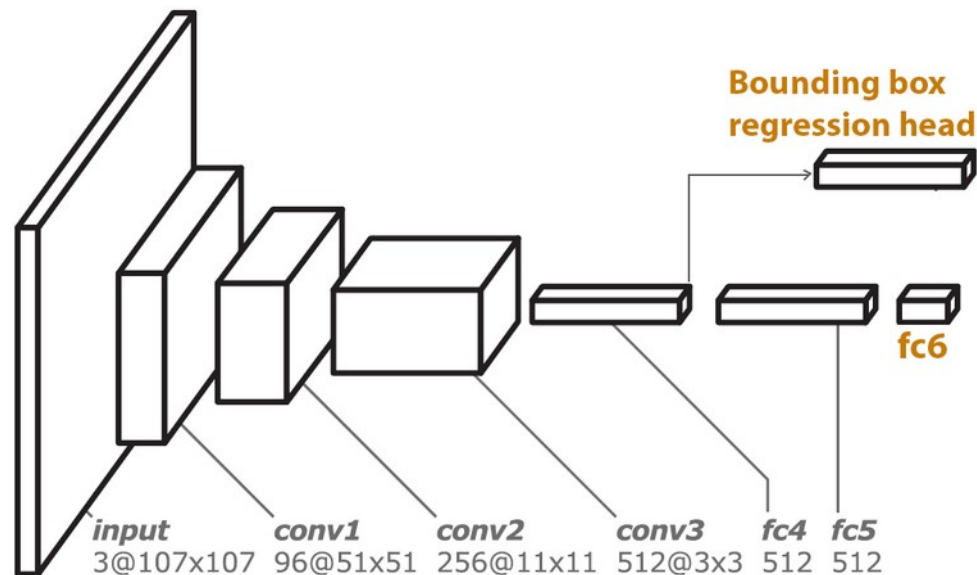


Deep learning for tracking

- Challenges
 - Efficient backbone pre-training
 - Training datasets
- Performance
 - Light-weight backbones
 - Efficient search
 - Online training

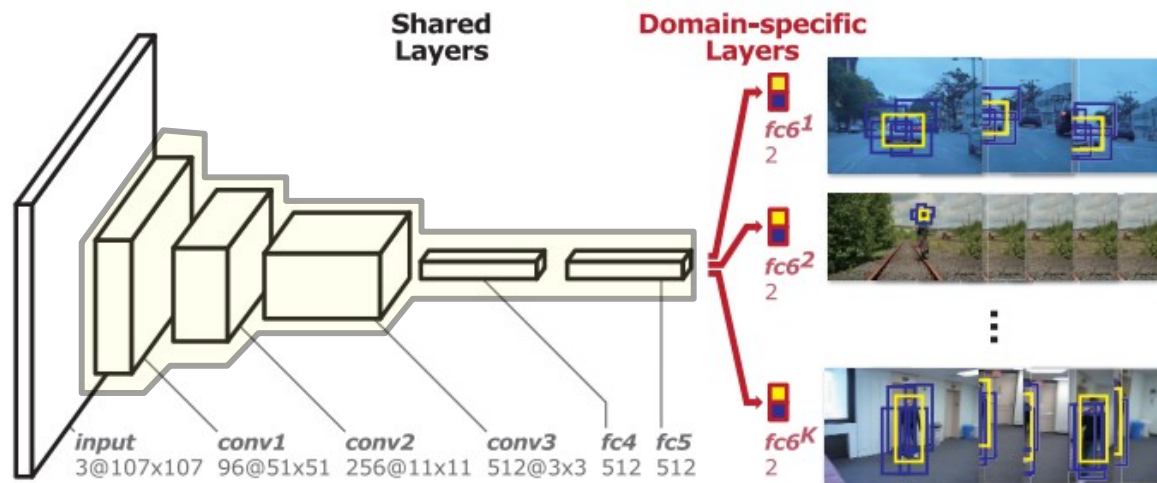
Case study: MDNet

- Multi-domain learning
- Extract and classify samples
- Conservative updates
- Hard negative mining
- Bounding-box regression
- Slow (1+ second per frame)

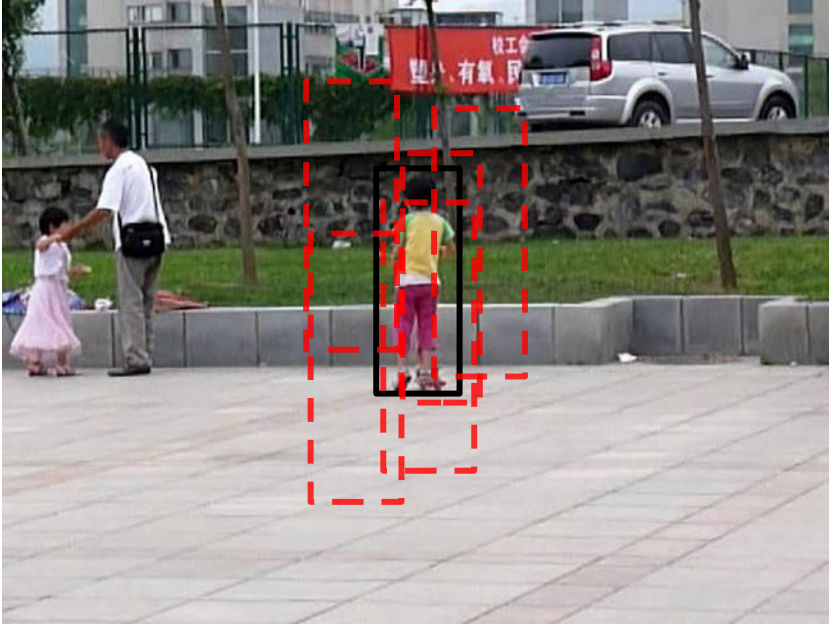


Backbone pre-training

- Pre-trained on training sequences
 - 50 positive, 200 negative samples per frame
 - Each sequence is a tracking domain
 - Shared backbone



Target localization



Sample bounding boxes
Compute classification scores



Take the box with max score
Regress the BB parameters

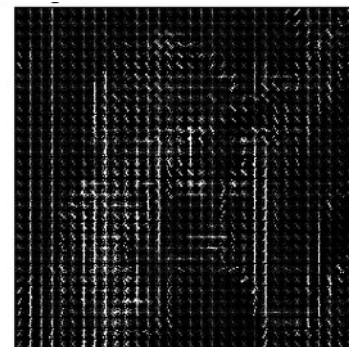
Initialization and updates

- Initialization
 - Fine-tune FC 4 and 5 layers train FC 6 from scratch
 - Train bounding box regression head
- Updates
 - Fine-tune all FC layers
 - Hard negative mining (difficult negative samples)
 - Short-term and long-term memory samples
 - Do not update during target loss



Back to correlation

- Intensity values not informative
 - Response not well expressed
 - Drift when appearance changes
- Histogram of Oriented gradients
 - Better, still low-level
 - No semantic properties

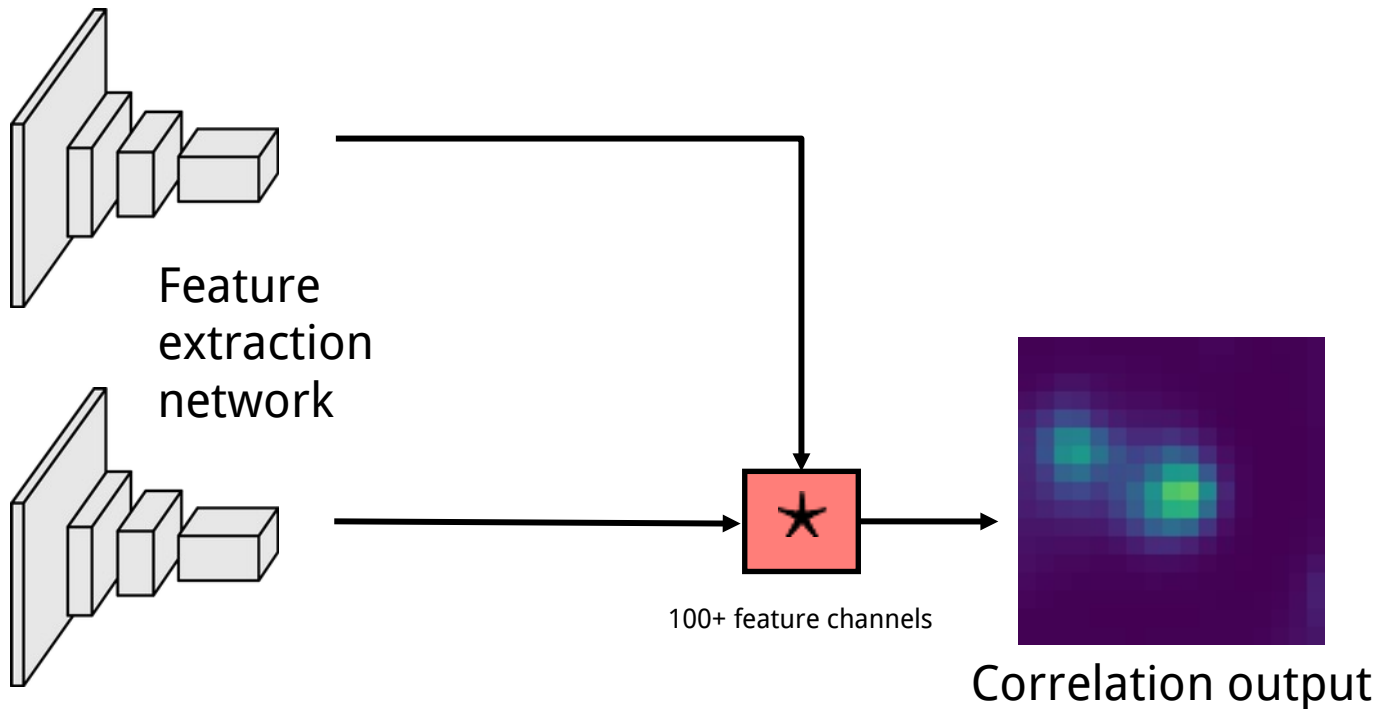
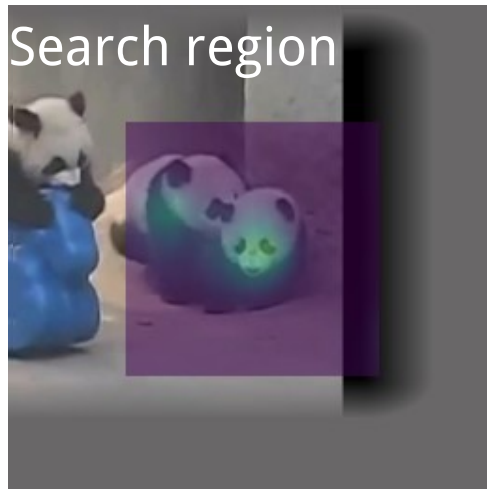


Correlation using deep features

Template



Search region



Correlation using deep features

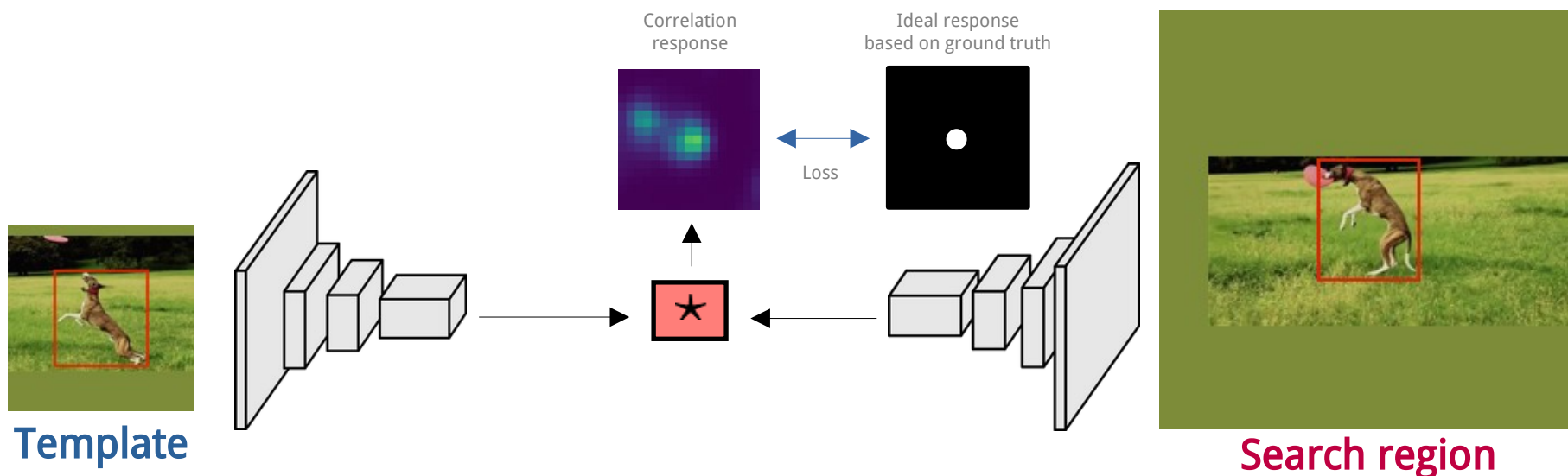
- Classification model
 - Convolution - local
 - Semantic properties
- Features were trained for classification
 - Not discriminative
 - Cannot distinguish between similar objects

Case study: SiamFC

- A fully convolutional Siamese network
- Template not updated
- Scale estimation with image pyramid
- 60fps (on GPU)

Pretraining

- ImageNet VID challenge
 - Video dataset with targets annotated
 - Pairs of random images from a sequence





Multiple objects?



Single-object

Generality

Different scenarios

Accuracy / Robustness



Multi-object

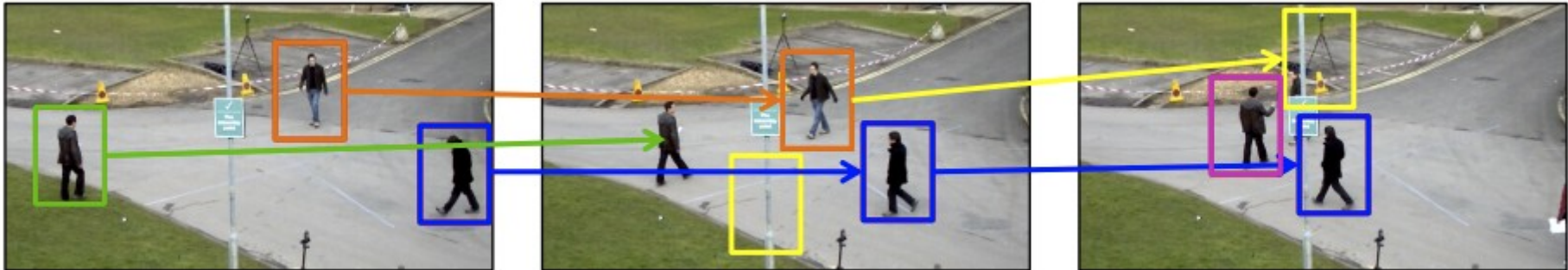
Category specific

Scenario specific

Re-identification

Detection and association

- Tracking-by-detection
 - Pre-trained category detector
 - Find detections that match and form a trajectory
 - Performance depends on the underlying methods



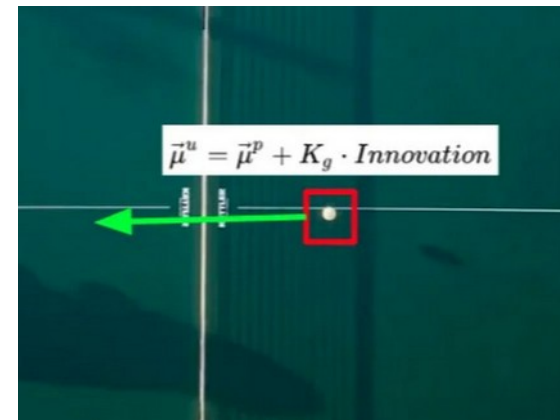
Appearance models

- Describe appearance
 - HoG, color
 - Deep embedding
- Retrieval problem
 - Cosine distance
 - Metric learning



Motion models

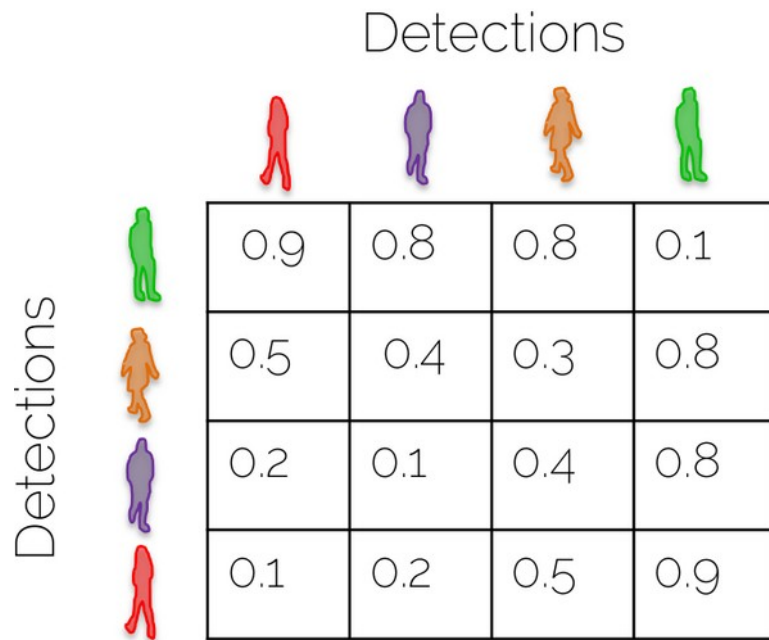
- Individual motion
 - Kalman filter
 - Nearly Constant Velocity
- Group motion
 - Social Force Models
 - Goal, Environment, Interaction
 - Pre-training



Frame-to-frame matching

- Obtain detections
- Define distances between detections
 - IoU, pixel distance
 - 3D distance
 - Appearance similarity
- Bipartite matching

Detections



				
	0.9	0.8	0.8	0.1
	0.5	0.4	0.3	0.8
	0.2	0.1	0.4	0.8
	0.1	0.2	0.5	0.9

Issues with locality

- Reliance on detector
 - If a detection is missing from a frame, we have to end the trajectory
 - Recovery is difficult
- Find the globally minimum cost solution
 - All frames
 - All objects/trajectories

Offline tracking

- Detections for the entire video
- Global optimization of graph
 - Nodes - detections
 - Edges - links
- Efficient occlusion handling

Summary

- Motion analysis done at multiple levels
 - Optical Flow
 - Object Tracking
- Object Tracking is a diverse field
 - Know your problem
 - Evaluate methods

Future directions

- Everything is **deep** nowadays
 - Optical Flow with large datasets
 - Tracking points over multiple frames
 - Tracking using Transformers
 - Tracking and Segmentation
 - Tracking multiple objects

Some recent development

- Using SAM for tracking (CVPR2025)



<https://jovanavidenovic.github.io/dam-4-sam/>